

Tab 1

Hypothesis progress update before memory architecture: [P3 Hypothesis Progress Update](#)

Running tally of experiments: [p3 team log - Google Docs](#)

White paper: [P3 Interim Report - Online LaTeX Editor Overleaf](#)

Results from conversation with superbuilder:

- Should have minimal AI writing (Mr. Liemandt takes notice of that)
- Use citations that directly relate and know exactly how they relate to the paper
- Absolutely avoid using jargon unless necessary, keep paper as terse as possible

Note: When doing a task, don't offload thinking/writing to AI (be able to explain what you did for task)

Tasks for 8/4/2026:

1. New approach: there's way too much AI slop on our doc to even sift through, so we're going to soft-reset (you can still copy-paste from the old white-paper with some caveats explained below).
2. Expectations:
 - a. The team responsible for a section has to write themselves a full explanation of what they did and how they did it (formatting is optional, academic language not needed but don't be loose/unclear with terms). Tables don't need to be included (but exact stats should be referenced if relevant), equations don't need to be cleanly written as long as they're intelligible/parseable, etc. This goes onto the staging tab for that category in this doc
 - b. One person per team is the migrator (uninformed on experiment): as a subteam finishes writing a section, read what's written, flag sections you don't understand/don't make sense or you think are slop
 - c. *Anything* on staging that hasn't been approved may be rejected, not guaranteed to go onto final report if it is insufficiently clear or slop
 - i. Explicitly: you will have to go back and either restart or modify a section of the staging category depending on review
 - d. After approval, the migrator formalizes it for putting onto final report (can use AI to "formalize" but it can't sound AI or restart, same ideas should be present). Make sure relevant citations are present, tables from old white paper are pasted if relevant, etc
 - i. Note: can start on migrating early/do it per section (might end up having to go back but will save time for the migrator later in the day), make minimal necessary set of changes while migrating
 - ii. If you're migrating and realize something is missing, subteam can add micro-update to staging for review without needing whole thing checked
3. Formulation staging: Eve (, Adarsh.ai), migrator: Claire
4. Shared arch staging: Adarsh, migrator: Andrew, William
5. "Standard regime" staging: Stephen, Yuen Kai, migrator: Ellie
6. Reasoning staging: Iris, Vishnu, migrator: Tejas
7. Info saturation staging: Eve, Preston, migrator: Anshul
8. Discussion, limitations, and future work (deferred):

Tasks for 8/3/2026 round 2 (check-in at 9):

1. Go through paper, any sections that look like human or that seem like they should be kept
 - a. Anshul, Andrew, Tejas, William
2. Delete section 3, only keep brief formulation
3. Read through our logs for info on the 2-hop reasoning task and the 160m run
 - a. Anshul, Andrew, Tejas, William
4. Do experiment 1 and redo experiment 1 design in doc
 - a. Vishnu
5. add results tables/images to doc section 5 and in other locations
 - a.
6. Fix paper formatting/structure
 - a. Eve, Iris
7. Remove/rework to talk about the goal of the experiment rather than relevant-but-besides-the-point notes
 - a.
8. Update citations
 - a. Note: Chinchilla, all the things in [] (ex one says [CITE]), the 2.5 bits/parameter figure are all things that need to be cited, but do a pass through on the references section and remove things that seem only tangentially relevant/add citations for other important figures used in paper
9. Continuing previous tasks
 - a. Claire, Ellie, Iris, Eve, Vishnu, Preston

Tasks for 8/3/2026 (check-in at 5):

1. Verifying validity of evals + Updating results in paper
 - a. Claire, Ellie
2. Fact saturation testing
 - a. Preston, Eve
3. Rewrite 160m results to de-emphasize lack of fact saturation, focus on standard training regime being unable to achieve fact saturation past 50k facts at 160million parameters, 8 billion training tokens (past Chinchilla optimality's 20:1, at ~50:1)
 - a. Exact stats about fact saturation already in AI-generated part of paper
 - b. Stephen, YC
- ~~4. Flagging white paper prose so it doesn't sound AI-generated (see note above)~~
 - a. Andrew
 - b. William
5. Fixing 2-hop results in white paper
 - a. Iris
6. Edit first 25% generally and write up split on multi-hop past 2 (new section 5.7)
 - a. Adarsh
- ~~7. Rework architecture (section 4) to integrate more with previous results, talk less about possible side effects and more on what we actually did~~

- a. Vishnu
- 8. Doing experiment 1 (math fact reasoning)
 - a. Vishnu
- ~~9. Experiment 1 methodology/setup/justification in white paper (results pending)~~
 - a. Anshul, Tejas
- 10. Any other task that has to be done (speak to me first)

White Paper Planning

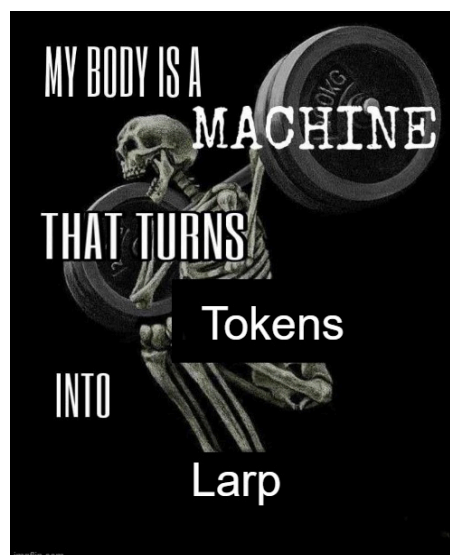
Summary: We're COOKED lol

- Two hop: Our model better at direct and two-hop by good amount, better at 2nd hop alone → preliminary sign that it generalizes across data types better (data access points act as separators)
 - Adarsh's is fried methodology lol, leave it out
- 160m: Our model does better on factual recall at all levels, but only ever comparable in reasoning
 - with held 8 billion tokens and 160million parameters (50:1 token-to-parameter ratio), #facts→#exposures goes 50k→196 200k→49 800k→12 so even well above Chinchilla unless you overtrain the model it will not meaningfully learn a wide array of facts
 - Reference exposures must be 100-1k
- Mathematical: no idea (up in the air)
- Fact saturation (Preston, Co-LMLM annotated data): no idea
- Fact saturation (Eve, super small models and synthetic saturation): no idea

Organization:

1. Intro + Background (multi-part thesis, how we want to address it, etc)
2. Related prior work (LMLM, Co-LMLM, their results)
3. Formulation (Define what counts as facts/reasoning, preexisting hypothesis, testable working theses)
4. Architecture/experiments (3-parted: prove split works for facts – significance, reasoning, normal regime limitation)
 - a. Possible 4th part: Fact Saturation?
5. Results/Discussion
 - a. Interpretation, Validity/Measurability
 - b. Things to cover: two-hop matches previously understood transformer difficulty with task; recall higher with standard regime and in two-hop = faster training
6. Conclusion
7. Appendix (finer details, citations)

Note: should we talk about fact externalization as a tool in SFT to stop unwanted access?



Old White Paper

On the Effects of Memory Splitting in Language Models

Lorem Ipsum
Department of Computer Science
Alpha University
XXXXX

Abstract

We investigate the benefits of externalizing factual knowledge from a language model. We trained a pair of models: one that learned the facts normally and one that never saw the facts and instead learned to query a key-value pair database for them. The split model does slightly better on reasoning (two-hop problems) and learns it faster. It hallucinates significantly less (roughly 0% vs 99%). [The split architecture is also beneficial in that it allows you to easily replace facts (eg. for different curricula) without having to retrain the entire model] -> NOTE: is this even true? we haven't really proven its better for reasoning. We also briefly investigate how effective teaching a dense model to query a database is for unlearning.

1 Introduction

Modern large language models (Vaswani et al, 2017) store facts and world knowledge in their parameters. Physics of Language models established that parametric knowledge occupies a finite budget. It appears that training a model to store more facts than the budget allows degrades recall, and finetuning to add or edit facts in weights interferes with previously learned skills, causing undesirable consequences such as factual staleness, hallucinations, and catastrophic forgetting.

This highly optimized repurposing of weights in LLMs is convenient but makes clean separation of internal mechanics difficult. Microsoft's Phi-3 technical report attributes the model's weak factual accuracy (e.g., on the TriviaQA dataset) to its limited capacity to store knowledge at small scale, and its authors deliberately filter highly specific facts from pretraining to "leave more model capacity for reasoning," recommending search-engine augmentation instead. Conversely, Deepseek v3 Engram reports gains not only on knowledge benchmarks but also on general reasoning (e.g., BBH + 5.0, ARC-Challenge + 3.7) by adding a constant-time lookup memory that offloads static pattern reconstruction.

SLOP i promise i wrote this by hand LMAOAOAOAO (sorry its not ai slop its just like we don't mention hippocampus anywhere after here :<) (wait youre valid i do have to integrate this better) (tyty this is <humannnnnnnn>)

In this experiment, we introduce *MemorySplit*, an alternative to parametric factual storage in model weights. In cognitive neuroscience, Complementary Learning Systems theory (CLS) holds that intelligent behavior relies on two interacting systems: a hippocampal system that encodes specific experiences and facts, and a neocortical system that distills generalizable structure. Operationalizing CLS, we train a dense transformer-based model to query *MemorySplit*, an external key-value database, for entity-level facts. During training, the retrieved fact values are masked from the loss, and at inference the database supplies the values. To isolate the effect of externalizing memory, we compare this model against an otherwise identical twin model trained to store the same facts (without *MemorySplit* implemented), holding data, order, and compute budget fixed.

2 Related Work

MemorySplit builds most directly on a recent line of work that externalizes factual memory during pre-training rather than at inference. [CITE] introduce Limited Memory Language Models (LMLM), whose pretraining recipe supports database lookup calls by masking retrieved facts from the loss, teaching a 1B-parameter model to query an external store rather than memorize facts in its weights. This differs from inference-time augmentation: methods such as nearest-neighbor language models (Knn-LM), RAG, and learned tool use leave the parametric copy of each fact intact and therefore does not make knowledge reliably editable or removable. We adopt the former loss-masking technique and study it in a controlled paired setting, presenting a mechanistic account of how the internal representation of *MemorySplit* differs from a vanilla model, as well as an evaluation of its effect on multi-step composition, other related reasoning benchmarks, and its potential utility for downstream AI safety tasks such as unlearning.

However, simply externalizing facts does not necessarily, by itself, improve reasoning. [CITE] show that models which retrieve stored facts perfectly can still fail to meaningfully manipulate them: [model name]’s performance on comparison and classification remain near chance without clear evidence of reasoning. More broadly, [Wang et al] show that reasoning tasks such as multi-hop composition fail to generalize to novel structure even after grokking, and such grokking itself emerges only after optimization far beyond the point of overfitting [THE ORIGINAL GROKKERS], [CHINCHILLA].

SLOP The black-box nature of an LLM’s internal workings, along with the ambiguity of where a fact differs from reasoning, makes their distinction difficult to test directly. Thus, we present the first controlled, mechanistic study in this area, assembling converging representational, causal, and behavioral evidence for how facts and reasoning interact when memory is externalized.

3 Formulation

Our main goal is to evaluate the qualitative and quantitative differences between a split-memory LLM and a dense LLM. One of the main claims we want to test is whether a model that does not store facts in its weights will reason better than one that does. Neither half of that sentence has a settled meaning in the literature, so we fix working definitions before describing any experiment. The definitions are narrow on purpose; they are chosen so that each hypothesis below is falsifiable by a measurement we can actually construct—and so that a negative result is informative rather than merely inconclusive.

3.1 Facts, storage, and retrieval

A *fact* is an atomic (entity, relation, value) triple. In our setting these are attributes of individual entities, such as a person’s birth city.

We separate two properties that are easily conflated. Let *Fact storage* be the ability to produce a value from the weights alone, with no external resource available and *fact retrieval* the ability to obtain that value by whatever means the architecture provides. For a conventional dense model the two coincide, since the weights are the only channel available. For a model that offloads values to an external store, storage is *theoretically* close to zero by construction while retrieval is supplied from outside. Storage and retrieval are therefore distinct properties, and the two architectures we compare differ on both at once.

This has a consequence for experimental design that we treat as binding. A comparison between a dense model answering closed-book and a split model answering with its store varies storage and retrieval together, so whatever difference it produces cannot be attributed to either. To separate them we need a third condition in which a dense model is supplied at evaluation with the same values the store supplies to the split model. Only the contrast against that condition speaks to reasoning; the contrast against closed-book dense measures the value of having the facts at all. We therefore treat the open-book dense condition as part of the minimal design rather than as an additional control.

Retrieval in the split architecture is not free, and we separate its two stages because they fail for different reasons and admit different fixes. *Addressing* is the model’s emission of a query key. *Resolution* is the store’s lookup of that key. Addressing is the learned capability, and it is the part that has to generalize to entities the model has not seen. Resolution is a property of the store and its coverage. We also distinguish two retrieval failures. An *abstention* occurs when the store cannot satisfy the query. A *silent error* occurs when a well-formed key resolves to a real but incorrect entry. These are not interchangeable, since the first is visible to a downstream system and the second is not.

One property of the store follows from the definitions rather than from engineering taste. If the store can traverse more than one relation in a single call, then it performs the composition that the model is supposed to perform, and the resulting comparison measures the retriever instead of the model. We therefore require that a single call returns the value for a single (entity, relation) pair, and that the store contains no edge that composes two relations. A store that violates this is a useful diagnostic but cannot serve as the main condition.

3.2 Reasoning

Reasoning is harder to pin down, and a single accuracy figure is a poor proxy for it. The same number can be produced by a memorized mapping, by a prior over answer formats, or by a scoring harness that fails on one class of outputs. Rather than define reasoning as a property of the model, which invites irresolvable philosophical debate, we define it more usefully as a property of the measurement:

An endpoint measures reasoning to the extent that, with fact retrieval and surface form held fixed,

1. accuracy declines as the gold derivation requires more serially dependent steps;
2. accuracy is unaffected by adding a premise that is topically plausible but logically irrelevant; and
3. accuracy collapses when a premise on which the derivation is based is removed.

Each part excludes a specific alternative account of a positive result. Depth sensitivity distinguishes derivation from recall of a stored answer. Invariance to an irrelevant premise distinguishes reading the premises from matching a surface pattern. Sensitivity to a removed premise establishes that the premises are load-bearing rather than decorative. None of the three can be satisfied by fact retrieval alone, which is what makes the criterion usable in a study whose treatment changes how facts are obtained.

We considered and rejected two weaker definitions. The first treats reasoning as whatever generalizes across many items whose answers do not appear in training, estimated through influence on the training set. This is appealing, but any task-wide shortcut satisfies it, and reasoning is generally fact/context-dependent [citation needed?]. The second treats reasoning as robustness to input perturbation (distractors) alone. Robustness is necessary but not sufficient, and it is also the weakest of the three parts above for small models, since a model performing near chance has no room to degrade.

The criterion carries a methodological requirement that we state here because it shapes every endpoint we later adopt. An endpoint is admissible only if we can state what a degenerate policy would score on it, what the chance rate is for the specific answer space rather than for the pool as a whole, whether the model’s output was parsed successfully, and how the score moves across several plausible phrasings of the same question. An endpoint lacking these cannot support a claim in either direction, because a null on it is indistinguishable from an instrument failure.

A further requirement follows from the storage and retrieval distinction. When a comparison is conditioned on whether the required facts were available, that condition must be defined from the corpus and the store, not from either model’s behavior on a probe. A probe conditioned on model output is not independent of the outcome it is being used to adjust.

3.3 Hypotheses

We test three claims, which stand in dependency order.

3.3.0.1 *H1: mechanism.*

Fact values can be relocated out of the weights. The split model’s closed-book recall and the fact content recoverable from its parameters both fall to near zero, while its recall with the store available is at least comparable to the dense model’s closed-book recall. H1 is a precondition rather than a result of interest in itself. If it fails, the treatment did not do what it was designed to do and the remaining hypotheses cannot be evaluated.

3.3.0.2 *H2: no capability cost.*

Relocating facts does not degrade general language ability. H2 is a guardrail. An apparent reasoning gain obtained by damaging the model in some other respect is not a gain, so H2 must hold for H3 to be worth interpreting.

3.3.0.3 *H3: the payoff.*

At matched parameter and token budgets, and measured on an admissible endpoint against the open-book dense condition, the split model reasons better than its dense twin.

H3 rests on a premise that we state separately because it is taken as fact by Liemandt et al., yet is supported nowhere in the literature [elaboration?]:

3.3.0.4 *H0: competition.*

Memorization and reasoning compete for limited capacity within the model, so decreasing one makes room for the other. Two things have to be true for this competition to be observable: the model must be operating near the limit of what it can store (Physics 3.3 claims 2 bits of information per parameter), and the reduction in stored content must be attributable to the treatment rather than to how often each fact appeared during training. If the premise does not hold in the regime we can afford to train in, then a null on H3 is evidence about the regime and not about the architecture, and we would rather be able to say which of those we have found.

We also record two secondary hypotheses, which are worth testing regardless of H3’s results.

3.3.0.5 H4: sample efficiency.

A model that is not required to memorize values reaches a given level of task competence in fewer optimization steps, even where its final accuracy is no better. This is a claim about the learning curve rather than its asymptote, and it is the form of benefit most relevant to training under a fixed compute budget.

3.3.0.6 H5: behavior on failure.

Because the split model is never trained to produce a value, it has no learned mechanism for inventing or fabricating one. We therefore expect its response to an unsatisfiable query to be abstention rather than hallucination, and we expect the knowledge it does rely on to be enumerable and editable without further training.

3.4 [PROBABLY CUT (SLOPPY)] What each outcome would establish

Because the definitions separate retrieval from reasoning, the possible results are individually interpretable, and we commit to reading them as follows. If the split model exceeds the closed-book dense model but matches the open-book dense model, the effect is fact retrieval: the architecture has changed where knowledge is kept without changing what the model does with it. If the split model exceeds the open-book dense model on an admissible endpoint, the effect is procedural and H3 is supported. If it falls below, H3 is falsified for the regime tested. We report the comparison against both dense conditions in every case, since reporting only the closed-book contrast would leave the first and second outcomes indistinguishable.

4 Architecture, Methodology, and Experiments

4.1 The paired dense–split design

We train six models: three dense/split pairs, one pair at each fact load $N \in \{n50k, n200k, n800k\}$ entities ($\approx 0.3M/1.2M/4.8M$ facts). Within each pair, the two models share an initialization seed, a data-order seed, a step count, and a token budget, and they are trained on a byte-identical token stream in the same order. In the style of LMLM (Zhao et al. 2025), the sole difference is the treatment of fact values: the **dense** arm sees unannotated inline text and is graded on the value tokens (it must memorize), while the **split** arm sees the same facts rewritten as a store lookup and has the value tokens masked from the loss (it learns only to issue the query). In this way, we reduce externalization of facts to the sole manipulated variable. We compare the two models’ performance on held-out evaluation sets.

[The backbone (d160m) is a decoder-only transformer with 12 layers, 12 heads, and $d_{model} = 768$ ($\approx 162M$ parameters): pre-norm RMSNorm, RoPE (base 10,000), a SwiGLU MLP of hidden width $round_{64}(\frac{8}{3}d_{model}) = 2048$, untied embeddings, no biases, causal attention, $N(0, 0.02)$ init with residual projections scaled by $1/\sqrt{2n_{layer}}$, context length 2048, vocabulary padded to 50,304. We optimize with AdamW ($\beta = (0.9, 0.95)$, weight decay 0.1 on rank- ≥ 2 parameters, gradient clip 1.0) under bf16, cosine schedule peaking at 1.5×10^{-3} after a 300-step warmup, decaying to a 10% floor; each of the $\approx 6.1k$ steps consumes 524,288 tokens (Table [tab:hparams]). Each arm sees $\approx 3.2B$ tokens over five components with fixed shares—FineWeb-Edu (0.54), synthetic biographies (0.23), iGSM-style arithmetic (0.12), RuleTaker-style deduction (0.08), fact-use QA (0.03)—with six attributes per biography rendered through many paraphrase templates.] -> feel like we can format this better

4.2 Part I: Externalization of facts

4.2.0.1 Why this comes first.

Every downstream claim about the split arm—no forgetting, editability, abstention, a reasoning comparison—is meaningful only if the split model truly holds no fact values in its weights. We therefore treat externalization as a hypothesis to be established, not assumed.

4.2.0.2 How the mechanism is built.

We extend the tokenizer with four control tokens ($\langle db_start \rangle$, $\langle db_retrieve \rangle$, $\langle db_end \rangle$, $\langle eot \rangle$). A fact is rendered $\langle db_start \rangle$ “name, relation” $\langle db_retrieve \rangle$ with loss on, the value with loss *off*, closed by $\langle db_end \rangle$; gradients flow through the query but never the value. The store is a trivial exact-match dictionary from a normalized “name, relation” key to a value; at inference the harness intercepts the decoded query and injects the retrieved value.

A training-time witness is the cross-entropy at masked fact positions in the split arm: it stays high, confirming the values are never learned. (This is a within-arm trace, not a cross-arm contrast—the dense corpus has no mask—and it should not be read against the uniform ceiling of 10.83 nats (=log padded vocabulary): a trained LM concentrates mass on frequent tokens, so at rare proper-noun positions it scores *worse* than uniform. The same probe on a dense arm that trained on those positions returns 25.03 nats with 98.7% of positions above the ceiling; a model that learned nothing scores ≥ 25 , whereas a split arm near 10.8 has learned the slot type but remains uncertain within the value pool—a weaker statement than never having learned the value.)

4.3 Part II: Effects on reasoning and two-hop composition

4.3.0.1 Task.

Each biography entity has atomic attributes (its bio facts) and, in the composition corpus, one bridge relation $r_1 \in \{\text{mentor}, \text{advisor}\}$ linking it to another entity. A two-hop item names a query person P and a target attribute r_2 (e.g. employer) and asks for that attribute of P 's bridge. Writing $B = r_1(P)$ for the bridge entity and $V = r_2(B)$ for its attribute, the answer is V , and reaching it requires composing two separately stored facts, $r_1(P) = B$ and $r_2(B) = V$. Neither the pair (P, V) nor the direct mapping $P \mapsto V$ ever appears in training, so the item cannot be answered by recalling a single memorized association.

We train the model to emit an explicit reasoning trace, the regime in which two-hop questions are reliably learnable (**wei2022cot?**; **press2023compgap?**): “Reasoning: P 's r_1 is B . Their r_2 is V . Answer: V ”. Composition thus decomposes into two learned sub-steps—recover the bridge B from P , then pose the second query about B . The second step is an *in-context copy*: the bridge B the model has just written (or looked up) must be carried into the r_2 query. It is not a memorized $P \mapsto V$ shortcut, and we do not presuppose a pre-existing copy/induction mechanism.

4.3.0.2 How each arm is trained.

The two arms see identical traces and, as in pre-training, differ only in how the two facts are rendered. In the **dense** arm every token is in the loss, so the model must recall both B and V from its weights (closed-book chaining). In the **split** arm each fact is a store lookup: the first hop is written $\langle \text{db_start} \rangle P, r_1 \langle \text{db_retrieve} \rangle B \langle \text{db_end} \rangle$, and the second, using that returned B , is $\langle \text{db_start} \rangle B, r_2 \langle \text{db_retrieve} \rangle V \langle \text{db_end} \rangle$. The returned values B and V are masked from the loss; the query strings are not. The split arm is therefore graded on *issuing* both lookups—including copying the store-returned bridge B into the second query—but never on producing B or V from its weights.

4.3.0.3 Populations and evaluation.

Following the synthetic-biography protocol of Allen-Zhu and Li (2023b) and its compositional extension (**compexposure2026?**), we split the entities into a *composed* population that appears in two-hop documents and a *held-out* population P_{held} that appears only in atomic single-hop biographies. We fine-tune the paired n50k bases on the composed population (10,000 entities, drawn from those the bases already store as facts), matched in data, order, and budget, and evaluate on three strata: (i) in-distribution composed pairs; (ii) P_{held} —held-out compositions of entities seen only as atomic facts, the out-of-distribution test of the two-hop *procedure*; and (iii) entities never seen at all. The split arm is evaluated with its store attached. To separate composition from fact access—the distinction at the heart of the compositionality gap (**press2023compgap?**)—the dense arm is evaluated both closed-book and open-book, with the two required facts supplied in context alongside distractors.

4.4 Part III: The limits of our regime

4.4.0.1 Why the regime is the story.

What a controlled memory-split study can conclude depends sharply on how saturated the models are, so we treat the training regime as an object of study in its own right rather than a fixed backdrop.

4.4.0.2 The fact-load dial.

We hold the biography budget fixed ($\approx 0.736\text{B}$ tokens) and vary the number of entities, which sets the per-fact exposure to ≈ 196 , 49, and 12 for n50k/n200k/n800k (Table 5); this dials memorization pressure without changing model size or compute. Each entity encodes ≈ 53 bits, so even the largest load (≈ 42 Mbit) sits well within the ≈ 324

Mbit parametric capacity at 2 bits/parameter (Allen-Zhu and Li 2024), and the models are trained at ≈ 20 tokens/parameter—roughly the compute-optimal (Chinchilla) budget (Hoffmann et al. 2022).

4.4.0.3 What we measure.

Dense storage versus exposures (bits/entity, the recall trajectory), weight occupancy versus capacity, the reasoning null and why it is uninformative here, and the forgetting/relearning differentiation between the arms (§[sec:exposure]).

5 Mechanistic Analysis

The split model holds its facts entirely outside its weights, while the dense model builds a compact, localizable mechanism to store and emit them. We summarize the headline contrasts in Table 1 and the causal-localization results in Table 2, and develop each thread below. All results are from a single training seed, so we read them as strong directions rather than proofs, and we are explicit about which are causal and which are correlational.

Mechanistic evidence that facts are externalized in the split arm (n50k, trained entities). Store-on/off recall, extractability, and the per-ability KL are behavioral; the position-decomposed membership-inference (MIA) AUC is statistical. Higher MIA AUC = the weights reveal which facts were in training; 0.5 is chance.

Measure	Dense	Split
Closed-book recall (store detached)	0.62	0.00
Recall with store attached	—	≈ 1.00
Extractability: generative recall / MC recognition	0.64 / 0.90	0.00 / chance
Position-decomposed MIA (AUC)	≈ 0.99	≈ 0.50
Per-ability KL between arms on value tokens (nats)	6.7 (vs ≤ 0.08 on all other abilities)	
Own top-64 fact-neuron selectivity ($ d $)	7–11	(dense units: 1.5–3 in split)

5.0.0.1 Behavioral.

[Mostly fine, wording’s a little weird at points] The split model recalls facts at $\approx 100\%$ with its store attached and *exactly* 0% with it detached, and a held-out likelihood analysis finds its weights maximally surprised by fact values ($\approx +7$ nats over dense [note: what does this measurement mean]) while both arms are equally fluent on knowledge-free text, showing that externalization carries no language-modeling tax. An extractability probe separates “never stored” from “stored but not producible”: dense at n50k is genuinely stored and producible (generative recall 0.64, multiple-choice recognition 0.90), whereas at higher loads and in the split arm both recall and recognition sit at the floor. Because the recognition test that succeeds on dense-n50k comes up empty elsewhere, the floor reflects real absence rather than a blunt instrument.

5.0.0.2 Statistical.

[SLOP] A position-decomposed membership-inference test [note: what does this mean] asks whether a model’s weights betray which facts were in its training set. The dense weights do so strongly (AUC ≈ 0.99), while the split weights are at chance (AUC ≈ 0.50): the split model’s parameters retain essentially no trace of the facts it was trained alongside. A per-ability distributional analysis [note: what does this mean] localizes the entire arm difference to the value tokens themselves. The two arms’ next-token distributions differ by ≈ 6.7 nats on biography values but by ≤ 0.08 nats on every other ability, so splitting changes what the model does at exactly the fact-value positions and nowhere else.

5.0.0.3 Causal.

[Maybe idk what this means but probably SLOP, way too many complex terms] Three interventions establish that dense builds fact machinery that is *necessary* for recall and *absent* in split (Table 2). First, ablating the 64 most fact-selective units in the dense model cuts its closed-book recall roughly in half ($0.66 \rightarrow 0.35$) against a $\approx 2\%$ drop for the same number of random units, and an alignment-free search, matching neurons by activation rather than index, finds no counterpart to these units anywhere in the split model. Second, mean-ablating a single late MLP (L9) at the value position more than halves dense recall ($0.61 \rightarrow 0.20$), whereas ablating a control layer (L6) barely moves it ($\rightarrow 0.53$), and the split arm has no closed-book recall to ablate. Third, direct logit attribution of the

committed value token tells the same story from the forward direction: dense’s value emission concentrates on the late MLPs (L11 dominant, then L10 and L9) and every dense prompt commits a value, whereas the split model commits a value on 0 of 100 prompts, leaving nothing to attribute. Finally, a dynamic relearning attack rules out “present but not linearly readable”: fine-tuning the split model to emit values recovers held-out recall no faster than fine-tuning it on entirely fresh, never-seen entities (both $\approx 0.3\%$ after 40 steps), while the dense model starts at 63%: a few gradient steps cannot surface values that were never stored. We report the layer ablations as localization evidence: the mean-ablation is blunt and not isolated from general late-layer function, so we do not claim a collateral-free intervention.

Causal localization on the twins (n50k, trained entities, database banned). Matched ablation: mean-ablating the dense late fact layer (L9) more than halves recall while a control layer (L6) does not; the split arm has no in-weight recall to ablate. DLA: fraction of prompts committing a value token at the value position. Relearn: closed-book recall after fine-tuning to emit values, at step 0 and step 40, with a fresh-entity control for split.

Intervention	Dense	Split
Fact-neuron ablation: recall (top-64 / random-64)	0.35 / 0.64	—
Matched MLP ablation: recall (baseline / L9 / L6)	0.61 / 0.20 / 0.53	0.00 / 0.00 / 0.00
DLA: prompts committing a value (of 100)	100	0
DLA: top contributing components	MLP L11, L10, L9	none
Relearn: recall (step 0 $\rightarrow$ 40)	0.63 $\rightarrow$ 0.48	0.00 $\rightarrow$ 0.003
split fresh-entity control (step 0 $\rightarrow$ 40)	—	0.00 $\rightarrow$ 0.003

image image

Causal localization on the twins (n50k, database banned). Left: mean-ablating the dense late fact layer (L9) more than halves closed-book recall while a control layer (L6) does not; the split arm has no in-weight recall to ablate. Right: direct logit attribution of the committed value token concentrates in dense’s late MLPs (L9–L11); the split model commits a value on 0/100 prompts, so there is nothing to attribute.

Relearning attack. Fine-tuning the split model to emit values recovers held-out recall no faster than fine-tuning it on fresh, never-seen entities (both ≈ 0), whereas the dense model starts near its stored level—the values were never latently present.

6 What/ Why we do the experiment [note: not sure if this heading makes sense]

§1 states the capacity argument this project tests. §5.1 settles two of its three parts: under loss masking the facts leave the weights entirely, with closed-book recall at zero and membership inference at chance, and offloading carries no language-modeling tax. The third part is whether the capacity so freed reaches reasoning, and the sweep of §4 and §5 could not close it.

[note: is this for prior runs/experiments? if so then we should probably state as such] Three obstacles stood in the way, each documented elsewhere in this paper. §5.2 shows the premise was never engaged, since occupancy peaked at 0.51% of the two-bit budget and nothing can be crowded out of a model that empty. §8.1 shows the reasoning endpoints failed independently of that, two of them scoring below their own constant-answer baselines. And §5.3 shows the one endpoint that cleared chance measured fact access together with a query-formation skill the split arm had rehearsed some 300,000 times and the dense arm never once.

[SLOP, also not sure what all the § means because there’s not §8.1 unless I’m not looking correctly] Formal proof removes all three at once. Lean 4 supplies the fact/reasoning boundary that §3 had to stipulate: a theorem’s statement is the fact, and the tactic script discharging it is the reasoning. A proof compiles or does not, so no constant policy scores above zero, which is the precise failure of §8.1. And the corpus below carries no premise citations, so there is no store to query and the skill transfer of §5.3 cannot arise.

[Note: strange wording] The question is therefore the project’s open one, asked in the cleanest setting we have found: at equal supervised tokens, is a model trained only on proofs a better prover than one trained on proofs and statements alike?

7 Setup

The paired design is §4’s: shared initialization, data order and token budget, differing only in the loss mask applied to fact spans. §4 documents three departures from that ideal, and this run closes two of them.

The token streams here are byte-identical. The split rendering adds no markup, so the corpus does not lengthen and the wrap that left dense at 1.006 passes against split’s 1.000 cannot occur. The token-accounting asymmetry is handled by running the split arm to two stopping points instead of one: A ends at the same total tokens as dense and serves as the undertraining control, while B extends the stream by $1/(1 - m) = 1.25$ so supervised tokens match, and carries the headline comparison [note: could be explained a bit better I feel, not sure what the next sentence means either]. A loss at A that becomes a tie at B is an artifact of counting. The third departure, that the arms are not scored on the same targets, is unchanged, and is why the endpoint is proof success rather than loss.

Budget	2B tokens, ~5.4 tokens per parameter
Optimizer	AdamW, 3e-4, weight decay 0.1, z-loss 1e-5
Batch	524,288 tokens per step, 16,384 per GPU, FSDP
Schedule	warmup 50M tokens, decay over the final 20%
Checkpoints	one every 250M tokens, 8 total

[SLOP] The budget sits below compute-optimal deliberately. §5.2 failed by leaving the dense arm too empty to contest, and saturating it would leave the split arm nothing to be relieved of; the contrast exists only while memorization is partial.

The endpoint is fixed before the run. Each arm receives 16 attempts per theorem at temperature 0.7, capped at 1,024 tokens, replayed against a pinned Lean toolchain under a 60-second timeout, and a theorem counts as solved if any attempt compiles. As in §5.1 there is one seed per arm, so noise is estimated within the run, from the last three checkpoints of each arm scored on a fixed 200-theorem subset. With Δ the split minus dense difference at stopping point B and $T = \max(3pp, 2\sigma)$, Δ above $+T$ is a win, within T a tie, and below $-T$ support for the null. A null additionally requires losing at both stopping points and passing two leakage probes: statement completion from a theorem’s name, checked against 50 planted canaries whose constants appear nowhere else, and a masked-versus-unseen perplexity gap at 2σ . [note: probably SLOP, this last paragraph is a little strange, especially the equations and the "canaries"]

8 Two-hop composition

8.1 Results

Both arms learn to compose on seen entities, and on that stratum the split arm’s advantage is fact *access*, not chaining (Table 7). Conditioned on both facts being accessible the dense model composes at 88.8%; its raw shortfall to the split arm traces to an $\approx 80\%$ access rate from lossy parametric memory rather than to a reasoning deficit. The split arm leaves a checkable trace—on aligned OOD items it emits the correct hop-1 and hop-2 lookup keys $\approx 99.8\%$ of the time, essentially equal to its answer accuracy—so its score comes from writing two correct queries and copying the bridge, not from a shortcut.

Two-hop composition after fine-tuning the n50k bases (chance $\approx 0.5\%$). Aligned entities were seen as facts by the base; novel entities never appeared in training. Dense is shown closed-book and “open” (the two facts supplied in context); split uses its store.

Condition	Aligned OOD	Novel OOD
Dense, closed-book	86.6	0.5
Dense, open (facts in context)	94.2	0.8
Split, external store	99.8	53.5

Split per-hop (aligned OOD): hop-1 key 99.9, hop-2 key 99.8, both 99.8

Split per-hop (novel): both keys 26 (single-hop 38–60; composes 83.7 when addressed)

Dense (aligned OOD): both-fact access rate 80.3; conditioned accuracy 88.8

8.2 Both arms perform genuine composition, not a shortcut

A causal test rules out a memorized person→answer shortcut in the dense arm. Overwriting the bridge entity the model itself wrote (teacher-forcing B_j in place of B_i) changes the final answer 100% of the time and makes it follow

the *injected* bridge’s own value 94.2% of the time, with a 0% shortcut rate; the $\approx 6\%$ residual is the injected bridge’s own unstable in-context recall, not shortcutting. The dense arm therefore computes hop 2 from the written bridge, which also explains why its two-hop accuracy exceeds the product of its cold single-hop recalls: hop 2 is an *in-context* read of the moved bridge, far more reliable than a context-free probe.

8.3 The composition circuits: dense vs split

The two arms solve the second hop with circuits of the same overall shape but different kind. In the **dense** arm, activation patching and a logit lens reveal a compact *retrieve* $\rightarrow$ *move* $\rightarrow$ *lookup* circuit: early layers build the written bridge’s representation on the bridge tokens; attention transports that identity to the final answer position (the bridge-token contribution collapses to zero by L9 while the final-position contribution rises to 1.0, crossing over around L7–8); and the attribute is looked up in the late layers—the answer is undecodable through L7 and snaps on at L8–9, fully formed by L9–11. The final answer position becomes a sufficient statistic for the whole computation by L9 (a single-site patch there recovers $\approx 100\%$ of the effect). In the **split** arm, the analogous late-layer machinery is a small control circuit—an L6 attention head that acts as a recursion driver, firing the *second* lookup, and an L11 MLP that copies the returned value into the composed answer. Both arms thus transport the bridge to the answer position and resolve hop 2 late (L8–L11), and both share a late copy/compose step; the essential difference is that dense performs an *in-weight attribute lookup* whereas split *emits and copies a second external query*—which is precisely why split needs the L6 recursion-driver that dense has no use for. We report these as converging localization evidence rather than a matched head-level comparison: the dense circuit was characterized on the composition checkpoint via position-level patching on self-elicited chains, and the split control circuit on its own fine-tuned run; a fully matched, component-level comparison on a single paired checkpoint is blocked by the composition store’s missing bridge edges and is left to future work. All circuit results are single-seed.

8.4 Sample efficiency and skill transfer

The split arm also converges $10\text{--}15 \times$ faster, reaching ceiling by $\approx 25\text{M}$ fine-tune tokens where the dense arm only begins composing at $\approx 100\text{M}$ and plateaus near 86% by $\approx 370\text{M}$. We attribute this to *skill transfer*, not freed capacity: during pre-training the split arm executed the primitive “read an entity name, copy it into a query slot, receive the value” across $\approx 300\text{k}$ keys at ≈ 196 exposures each, with loss on for the query-formation tokens, whereas the dense arm executed it zero times. Split therefore enters the fine-tune with the copy-to-slot circuit already installed and needs only to learn the bridge, while dense must learn both. This addressing skill generalizes to unseen *synthetic* entities ($\approx 100\%$ correct lookup keys) but is distribution-bound: on real-world PopQA entities the lookup rarely fires and never copies a real name.

8.5 Caveats

Two caveats bound the novel-entity column, and both cut against reading it as novel-composition generalization. First, the dense arm was never trained to consume retrieved facts, so its open-book evaluation is zero-shot and out-of-format, making its 0.8% plausibly a formatting rather than a binding failure. Second, the biography generator draws attributes from fixed pools, so reseeding produces new *names* but not new *values*: the regenerated novel population has 0.0% name overlap with the base yet 100% attribute-value overlap. Only the first hop is therefore out of distribution, so the 53.5 versus 0.8 result is a first-hop *access* gap rather than novel composition, consistent with the first-hop bottleneck reported for latent multi-hop reasoning (Yang et al. 2024). All results are single-seed and should be read as directions.

9 Exposures

9.1 Fact saturation under normal training regimes

Split does not improve reasoning, because under normal training regimes the model never overcrowds.

At 50k entities, the dense model holds around 33 of the roughly 53 bits per entity that we showed it, but at 200k and 800k it holds about 0.2 bits per entity, which is close to nothing. What predicts this is not the number of facts but the number of times the model sees each one, which falls from about 196 exposures to 49 and then 12 as the load rises. A recognition probe confirms the absence, as the same probe reads 0.90 where storage exists and sits at chance everywhere else. According to Physics of Language Models, a 162M model has room for roughly 324M bits of factual content. At our highest load, we asked it to store about 13 percent of that and it retained around 0.05 percent.

Across the whole sweep, it was never more than half a percent full, so it seems unlikely that crowding could be doing much of anything.

Reading the dense column of our results the other way is also informative. Going from 50k to 800k entities multiplies the fact load by sixteen, and dense reasoning accuracy moves from 65.0 to 66.7 to 68.2, so it does not fall. The crowding account predicts degradation, but we see at least a flat reasoning, discrediting the crowding account. We assumed a model shown too many facts would compress them until something gave way, whereas what it actually seems to do is decline to learn them, and that costs it nothing.

Lowkey slop that needs relevant factoids kept:

- Model size: 162, 220, 800 parameters.
- Entity loads: 50, 000, 200, 000, and 800, 000.
- Estimated information per entity: approximately 53 bits.
- Estimated factual capacity at two bits per parameter: approximately 324 Mbit.
- At 800, 000 entities, factual demand was 42. 4 Mbit, or 13. 1% of estimated capacity.
- Exposures per entity were approximately 196, 49, and 12 at loads of 50, 000, 200, 000, and 800, 000, respectively.
- Split-minus-dense held-out deduction scores were $- 2. 0$, $+ 3. 1$, and $- 5. 5$ points at the three respective loads.
- The difference-in-differences from 50, 000 to 800, 000 entities was $- 3. 5$ points.
- Dense-model storage was approximately 33 bits per entity at 50, 000 entities and 0. 2 bits per entity at both 200, 000 and 800, 000 entities.
- The recognition probe scored 0. 90 where storage was detected and remained at chance level elsewhere.
- Estimated capacity occupied at 800, 000 entities was approximately 0. 05%.
- Estimated capacity occupied anywhere in the sweep did not exceed approximately 0. 5%.
- Dense held-out deduction accuracy was 65. 0, 66. 7, and 68. 2 at the three respective loads.
- Entity load increased by a factor of 16 from 50, 000 to 800, 000.
- Entity count and exposures per entity were not varied independently.
- The referenced endpoint evaluations are no longer available.
- The referenced compute-optimal training estimate was approximately 20 tokens per parameter.

9.2 The capacity premise was never engaged

The hypothesis this experiment set out to test requires facts to crowd the weights until reasoning suffers, and the corpus we built cannot produce that pressure. Its largest load carries 800,000 entities at roughly 53 bits each, which is 42.4 Mbit of factual demand against the 324 Mbit that 162,220,800 parameters afford at two bits per parameter. The corpus therefore makes only 13.1 percent of the capacity budget available to be contested. That is before any question of whether the model stores what it is offered. Exposures compound the problem, falling from roughly 196 per entity at the smallest load to 49 and then 12 at the largest. Those counts are far below what parametric memorization is reported to need. Therefore the premise was inactive by construction, and no endpoint measured on this sweep could have adjudicated the hypothesis.

We state that as a design result rather than an empirical one, because it follows from the corpus arithmetic and requires no evaluation. This matters, since the evaluations that would have supplied the endpoint no longer exist. Compute-optimal training at roughly 20 tokens per parameter falls well short of the per-fact exposure that memorization requires. The grokking literature places compositional generalization only after loss saturation, which points the same way. A run in this regime is undertrained for the phenomenon it intends to measure, and that is the most useful thing this sweep establishes.

9.3 The exposure question

Our experiments are clearly leaning towards a null hypothesis. On held-out deduction, the split arm scores 2.0 points below its dense twin at 50k entities, 3.1 above at 200k, and 5.5 below at 800k. There is no consistent advantage, and

the sign reverses across the dose. The hypothesis predicts that the advantage should grow as fact load grows, since that is when the dense twin should be most burdened, and instead the difference in differences comes out at negative 3.5 points.

The storage measurements explain why we see these results. At 50k entities, the dense model holds around 33 of the roughly 53 bits per entity that we showed it, but at 200k and 800k it holds about 0.2 bits per entity, which is close to nothing. What predicts this is not the number of facts but the number of times the model sees each one, which falls from about 196 exposures to 49 and then 12 as the load rises. A recognition probe confirms the absence, as the same probe reads 0.90 where storage exists and sits at chance everywhere else. By Knowledge Capacity Scaling Laws, a 162M model has room for roughly 324M bits of factual content (Allen-Zhu and Li, 2024). At our highest load, we asked it to store about 13 percent of that and it retained around 0.05 percent. Across the whole sweep, it was never more than half a percent full, so it seems unlikely that crowding could be doing much of anything. Reading the dense column of our results the other way is also informative. Going from 50k to 800k entities multiplies the fact load by sixteen, and dense reasoning accuracy moves from 65.0 to 66.7 to 68.2, so it does not fall. The crowding account predicts degradation, but we see at least a flat reasoning, discrediting the crowding account. We assumed a model shown too many facts would compress them until something gave way, whereas what it actually seems to do is decline to learn them, and that costs it nothing.

We are not claiming the null hypothesis holds everywhere. The conjecture still looks testable in a regime where models genuinely do memorise, and our data place that boundary somewhere between 49 and 196 exposures per fact, which means we should aim for 200 exposures per fact in an experiment. Our biography budget was limited, so raising the entity count automatically lowered the exposures per entity, and the two moved together in our sweep. Separating them is the point of the experiment we would like to run next, which holds exposures at around 200 and scales the token budget with the entity count rather than against it. That is the one regime where the dense arm carries a real memorisation burden, and so it is the only place where freed capacity would refer to something that exists.

9.4 Abstention and editability, as design properties rather than measurements

An earlier version reported a hallucination and leakage audit over 5,000 trials per cell. In it the dense arm fabricated a plausible wrong value for an unseen person roughly 99 percent of the time while the split arm did so essentially never and leaked no value with its store detached. We withdraw those figures for the same reason as the rest of this section, which is that no artifact in the project produces them.

Two claims survive the withdrawal, because they follow from the architecture rather than from a measurement. A value the split arm was never trained to produce cannot be revoked from its weights, so deletion reduces to removing a row from the store. A curriculum can likewise be swapped without touching the model. Whether the model’s behavior after revocation is abstention or confabulation is an empirical question, and it is exactly the question the withdrawn audit was built to answer. We flag it as the most promising thing to measure first when the evaluation harness is rebuilt. The abstention result was the sharpest effect the earlier draft reported, and it may well survive a rerun.

Held-out fact-use question answering at the 200k load, 0.8B-token pilot, by question family. Accuracies are percentages and the last column gives each family’s contribution in points to the overall difference. The same-city row is a constant response in both arms, and the extraction row is at floor in both.

Question family	items	dense	split	share of gap
Date comparison	598	45.7	96.5	+ 20.3
Same-city check	481	40.5	59.5	+ 6.1
Attribute extraction	421	0.0	2.1	+ 0.6
All	1,500	31.2	58.1	+ 26.9

9.5 Real-KB scale-up: subject realization limits exact retrieval

We tested whether the exact-store interface transfers from synthetic names to open-vocabulary Wikidata5M entities. Each evaluation item asks the split model to emit one or more textual lookups of the form subject, relation. A lookup succeeds only when both fields match a stored key after lowercase and whitespace normalization.

The main failure occurs in the subject field. Of 4,735 evaluation items, 4,654 produced a scorer-parsed first lookup. The first relation ID matched the gold relation in 55.29 percent of those items, compared with a 28.64 percent

post-hoc constant-relation diagnostic over ten relation IDs. The exact gold [note: what is a gold subject] subject surface appeared in only 1.68 percent. Because the first-hop subject is present verbatim in the question, this is a copy-to-query-slot failure under the executed exact-surface interface, not evidence that the subject was absent from the prompt.

The same pattern persists over the full decoded output. Across 23,687 scorer-parsed lookup strings, relation-ID accuracy was 35.35 percent and exact subject accuracy was 0.48 percent. Generation-time instrumentation separately recorded 23,607 served lookups, with exact-store hit rates between 2.0 and 3.1 percent across evaluation slices. The scorer-parsed and generator-state events do not align one-for-one, so these measurements do not identify a wrong-row rate.

The model also queried far more often than the gold paths required. It issued an average of 4.99 served lookups per item, while the mean gold path contained 1.44 hops. This excess reflects repeated attempts after misses and is not due to error propagation across later hops.

The run used a 300,003-entity Wikidata5M source graph, a 50,000-entity training dose, and a 275,973-edge organizer. A 356M-parameter dense model and split model were each trained for 7.20B tokens at seed 0. Both arms fell below the best post-hoc constant-answer policy on every full evaluation slice, so we do not interpret their accuracy difference [not sure what this part means and it sounds kinda like nothing worked]. The comparison was also exposure-confounded: the shorter dense rendering produced 22.6 percent more clean multi-hop document occurrences and 59.6 percent more requery occurrences. Equal cumulative tokens therefore did not provide an equal training comparison; paired runs must match document occurrences as well as token budgets [note: not sure what this means, does that mean some part of this experiment is highly confounded? if it's an error in a previous run, why include it].

The synthetic gate results and real-graph result answer different questions. The synthetic split checkpoint produced 3,953 well-formed calls with no malformed queries, showing that query syntax can generalize within a closed pool of canonical names. In the real-graph run, 98.3 percent of items still produced a parseable first lookup, but the exact subject rarely denoted the required row. Syntax therefore transfers much more reliably than reference.

The held-out-phrasing slice tests only one reserved template and should not be read as broad linguistic generalization. Its entity-structure pairs are disjoint from training, but 597 of 760 items reuse at least one training edge, and 120 of 141 three-hop items share the same gold answer. These overlaps make the slice a narrow test of template transfer rather than a general test of unseen composition [note: this also seems like a weakpoint in the experiment, a lot of this subsection also just seems like talking about how the experiment didn't work].

Trace-preserving checkpoint reruns reproduced every archived store-on outcome exactly and produced byte-identical evaluation files on A100 and A10G GPUs. The result is therefore reproducible for the final checkpoint: exact open-vocabulary subject realization is the main constraint for execution, while relation selection, retry efficiency, and the reasoning-capacity hypothesis remain unresolved [note: not sure what any of these hypotheses are].

10 Discussion

SLOP

SLOP

11 Limitations and Future Work

Our study is bounded in ways that its own analysis makes precise. It uses a single architecture and largely a single seed, so cross-arm differences are directions rather than proven effects; it uses a synthetic factual corpus whose fixed attribute pools create the first-hop/second-hop asymmetry noted above; and, most consequentially, it runs at a compute-optimal budget that is exposure-starved for memorization and pre-saturation for grokking, so the capacity-crowding hypothesis is untested rather than falsified. The natural next step is to re-run the comparison in a regime where the premise is engaged: more exposures or a larger fact-to-capacity ratio, at which point a genuine reasoning gap, if one exists, should become measurable. Separately, the abstention and editability results motivate a dedicated line of work on the external store as a safety instrument: using MemorySplit-style offloading for targeted unlearning, curriculum swapping, and privacy revocation, and comparing it against parametric unlearning baselines. We view this safety application, rather than reasoning, as the most promising direction the present results open.

12 Experimental Validation

The fact-load dial. Because the biography token budget is held fixed, entity count and exposures per entity are perfectly anti-correlated—a confound we return to in Section [sec:160m-limits].

Setting	Entities	Facts	Exposures per entity
n50k	50,000	0.3M	≈ 196
n200k	200,000	1.2M	≈ 49
n800k	800,000	4.8M	≈ 12

Dense parametric storage as a function of exposures per fact. The information-theoretic ceiling is approximately 53 bits per entity by construction. Below the threshold the model declines to learn rather than compressing.

Setting	Exposures	Stored bits/entity	Regime
n50k	≈ 196	≈ 33.1	Memorization active
n200k	≈ 49	≈ 0.20	Collapsed
n800k	≈ 12	≈ 0.20	Collapsed

Two-hop composition accuracy after fine-tuning from the n50k bases. Chance is approximately 0.5%. Aligned entities appear in the base’s own training set; novel entities were never seen.

Condition	Aligned	Novel
Dense, closed-book	86.6	0.5
Dense, open (facts in context)	94.2	0.8
Split, external store	99.8	53.5

Hallucination and leakage audit across fact loads. Bounds in parentheses are the one-sided upper limits at $n = 5,000$.

Metric	n50k	n200k	n800k
<i>Dense, closed-book</i>			
Unseen person → fabricates	99.4%	99.5%	99.1%
real-but-wrong value			
Unseen person → correct by chance	0.7%	0.5%	0.7%
Trained person → correct recall	60.8%	0.7%	0.9%
Trained person → fabrication	39.0%	98.8%	98.0%
<i>Split, external store</i>			
Store on, key present → correct retrieval	100%	98.0%	99.5%
Store on, miss → fabricates a value	0.00%	0.00%	0.08%
Store on, miss → retries the query	45.9%	37.0%	30.4%
Store on, miss → blank	0%	0%	0%
Store detached → leakage from weights	0/2000	0/2000	0/2000
Store detached → fabrication	0/2000	0/2000	2/2000
Invariant violation (value emitted without store)	0 ($\leq 0.077\%$)	0 ($\leq 0.077\%$)	4 ($\leq 0.206\%$)
Wrong-row collision (real value, wrong person)	0 ($\leq 0.077\%$)	0 ($\leq 0.077\%$)	0 ($\leq 0.077\%$)

Appendices

- Abdin, Marah, Jyoti Aneja, Hany Awadalla, et al. 2024. “Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone.” *arXiv Preprint arXiv:2404.14219*.
- Abdin, Marah, Jyoti Aneja, Harkirat Behl, et al. 2024. “Phi-4 Technical Report.” *arXiv Preprint arXiv:2412.08905*.
- Allen-Zhu, Zeyuan. 2025. “Physics of Language Models: Part 4.1, Architecture Design and the Magic of Canon Layers.” *arXiv Preprint arXiv:2512.17351*.
- Allen-Zhu, Zeyuan, and Yuanzhi Li. 2023a. “Physics of Language Models: Part 3.1, Knowledge Storage and Extraction.” *arXiv Preprint arXiv:2309.14316*.
- Allen-Zhu, Zeyuan, and Yuanzhi Li. 2023b. “Physics of Language Models: Part 3.2, Knowledge Manipulation.” *arXiv Preprint arXiv:2309.14402*.
- Allen-Zhu, Zeyuan, and Yuanzhi Li. 2024. “Physics of Language Models: Part 3.3, Knowledge Capacity Scaling Laws.” *arXiv Preprint arXiv:2404.05405*.
- Barron, Joshua, and Devin White. 2025. “Too Big to Think: Capacity, Memorization, and Generalization in Pre-Trained Transformers.” *arXiv Preprint arXiv:2506.09099*.
- Biderman, Stella, Hailey Schoelkopf, Quentin Anthony, et al. 2023. “Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling.” *International Conference on Machine Learning (ICML)*.
- Cao, Shulin, Jiaxin Shi, Liangming Pan, et al. 2022. “KQA Pro: A Dataset with Explicit Compositional Programs for Complex Question Answering over Knowledge Base.” *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 6101–19.
- Carlini, Nicholas, Steve Chien, Milad Nasr, Shuang Song, Andreas Terzis, and Florian Tramer. 2022. “Membership Inference Attacks from First Principles.” *IEEE Symposium on Security and Privacy (s&p)*, 1897–914.
- Carlini, Nicholas, Florian Tramer, Eric Wallace, et al. 2021. “Extracting Training Data from Large Language Models.” *30th USENIX Security Symposium (USENIX Security)*, 2633–50.
- Chen, Mark, Jerry Tworek, Heewoo Jun, et al. 2021. “Evaluating Large Language Models Trained on Code.” *arXiv Preprint arXiv:2107.03374*.
- Chen, Qi, Bing Zhao, Haidong Wang, et al. 2021. “SPANN: Highly-Efficient Billion-Scale Approximate Nearest Neighbor Search.” *Advances in Neural Information Processing Systems (NeurIPS)*.
- Chen, Xiaoyi, Haoyuan Wang, Siyuan Tang, et al. 2026. “PrivUn: Unveiling Latent Ripple Effects and Shallow Forgetting in Privacy Unlearning.” *arXiv Preprint arXiv:2604.22076*.
- Cheng, Xin, Rui Tian, Wangding Zeng, et al. 2026. “Conditional Memory via Scalable Lookup: A New Axis of Sparsity for Large Language Models.” *arXiv Preprint arXiv:2601.07372*.
- Clark, Peter, Isaac Cowhey, Oren Etzioni, et al. 2018. “Think You Have Solved Question Answering? Try ARC, the AI2 Reasoning Challenge.” *arXiv Preprint arXiv:1803.05457*.
- Clark, Peter, Oyvind Tafjord, and Kyle Richardson. 2020. “Transformers as Soft Reasoners over Language.” *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI)*, 3882–90.
- Cobbe, Karl, Vineet Kosaraju, Mohammad Bavarian, et al. 2021. “Training Verifiers to Solve Math Word Problems.” *arXiv Preprint arXiv:2110.14168*.
- Diao, Shizhe, Yu Yang, Yonggan Fu, et al. 2025. “Nemotron-CLIMB: CLustering-Based Iterative Data Mixture Bootstrapping for Language Model Pre-Training.” *arXiv Preprint arXiv:2504.13161*.
- ElSahar, Hady, Pavlos Vougiouklis, Arslan Remaci, et al. 2018. “T-REx: A Large Scale Alignment of Natural Language with Knowledge Base Triples.” *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC)*.
- European Parliament and Council of the European Union. 2016. *Regulation (EU) 2016/679 of the European Parliament and of the Council (General Data Protection Regulation)*. Official Journal of the European Union L 119/1.
- Farooq, Ahmed. 2025. “A Framework for Non-Linear Attention via Modern Hopfield Networks.” *arXiv Preprint arXiv:2506.11043*.

Feldman, Yair, Linxi Zhao, Nathan Godey, et al. 2026. “Co-LMLM: Continuous-Query Limited Memory Language Models.” *arXiv Preprint arXiv:2607.07707*.

Gemma Team. 2024. “Gemma: Open Models Based on Gemini Research and Technology.” *arXiv Preprint arXiv:2403.08295*.

Grover, Aditya, and Jure Leskovec. 2016. “Node2vec: Scalable Feature Learning for Networks.” *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 855–64.

Hendrycks, Dan, Collin Burns, Steven Basart, et al. 2021. “Measuring Massive Multitask Language Understanding.” *International Conference on Learning Representations (ICLR)*.

Hendrycks, Dan, Collin Burns, Saurav Kadavath, et al. 2021. “Measuring Mathematical Problem Solving with the MATH Dataset.” *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*.

Hoffmann, Jordan, Sebastian Borgeaud, Arthur Mensch, et al. 2022. “Training Compute-Optimal Large Language Models.” *Advances in Neural Information Processing Systems (NeurIPS)*.

Hopfield, John J. 1982. “Neural Networks and Physical Systems with Emergent Collective Computational Abilities.” *Proceedings of the National Academy of Sciences* 79 (8): 2554–58.

Jégou, Hervé, Matthijs Douze, and Cordelia Schmid. 2011. “Product Quantization for Nearest Neighbor Search.” *IEEE Transactions on Pattern Analysis and Machine Intelligence* 33 (1): 117–28.

Jelassi, Samy, Clara Mohri, David Brandfonbrener, et al. 2024. “Mixture of Parrots: Experts Improve Memorization More Than Reasoning.” *arXiv Preprint arXiv:2410.19034*.

Jiang, Albert Q., Alexandre Sablayrolles, Arthur Mensch, et al. 2023. “Mistral 7B.” *arXiv Preprint arXiv:2310.06825*.

Johnson, Jeff, Matthijs Douze, and Hervé Jégou. 2021. “Billion-Scale Similarity Search with GPUs.” *IEEE Transactions on Big Data* 7 (3): 535–47.

Johnston, David, and Nora Belrose. 2025. “Examining Two Hop Reasoning Through Information Content Scaling.” *arXiv Preprint arXiv:2502.03490*.

Joshi, Mandar, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. 2017. “TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension.” *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*, 1601–11.

Jun, James P., Vijay Marupudi, Raj Sanjay Shah, and Sashank Varma. 2025. “A Neural Network Model of Complementary Learning Systems: Pattern Separation and Completion for Continual Learning.” *Proceedings of the Annual Meeting of the Cognitive Science Society (CogSci)*.

Karmim, Yannis, Luis Marti, Djamé Seddah, and Valentin Barrière. 2026. “Multi-Hop Knowledge Composition Is Bound by Pretraining Exposure.” *arXiv Preprint arXiv:2606.09338*.

Kashyap, Satyananda, Niharika S. D’Souza, Luyao Shi, Ken C. L. Wong, Hongzhi Wang, and Tanveer Syeda-Mahmood. 2024. “Modern Hopfield Networks Meet Encoded Neural Representations – Addressing Practical Considerations.” *arXiv Preprint arXiv:2409.16408*.

Kingma, Diederik P., and Max Welling. 2014. “Auto-Encoding Variational Bayes.” *International Conference on Learning Representations (ICLR)*.

Kornblith, Simon, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. “Similarity of Neural Network Representations Revisited.” *International Conference on Machine Learning (ICML)*, 3519–29.

Lewis, Mike, Yinhan Liu, Naman Goyal, et al. 2020. “BART: Denoising Sequence-to-Sequence Pre-Training for Natural Language Generation, Translation, and Comprehension.” *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 7871–80.

Lewis, Patrick, Ethan Perez, Aleksandra Piktus, et al. 2020. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.” *Advances in Neural Information Processing Systems (NeurIPS)*.

Loshchilov, Ilya, and Frank Hutter. 2017. “SGDR: Stochastic Gradient Descent with Warm Restarts.” *International Conference on Learning Representations (ICLR)*.

Loshchilov, Ilya, and Frank Hutter. 2019. “Decoupled Weight Decay Regularization.” *International Conference on Learning Representations (ICLR)*.

Maini, Pratyush, Zhili Feng, Avi Schwarzschild, Zachary C. Lipton, and J. Zico Kolter. 2024. “TOFU: A Task of Fictitious Unlearning for LLMs.” *Conference on Language Modeling (COLM)*.

Mallen, Alex, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. “When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories.” *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 9802–22.

Marek, Martin, Dongkyu Cho, Shikai Qiu, Rumi Chunara, Pavel Izmailov, and Andrew Gordon Wilson. 2026. “Forgetting in Language Models: Capacity, Optimization, and Self-Generated Replay.” *arXiv Preprint arXiv:2605.26097*.

Martins, André F. T., and Ramón Fernandez Astudillo. 2016. “From Softmax to Sparsemax: A Sparse Model of Attention and Multi-Label Classification.” *International Conference on Machine Learning (ICML)*, 1614–23.

Masumura, Tsubasa, and Masato Taki. 2025. “On the Role of Hidden States of Modern Hopfield Network in Transformer.” *Advances in Neural Information Processing Systems (NeurIPS)*.

Meng, Yu, Jiaxin Huang, Guangyuan Wang, et al. 2019. “Spherical Text Embedding.” *Advances in Neural Information Processing Systems (NeurIPS)*.

Mikolov, Tomas, Ilya Sutskever, Kai Chen, Greg S. Corrado, and Jeff Dean. 2013. “Distributed Representations of Words and Phrases and Their Compositionality.” *Advances in Neural Information Processing Systems (NeurIPS)* 26.

Morris, John X., Chawin Sitawarin, Chuan Guo, et al. 2025. “How Much Do Language Models Memorize?” *arXiv Preprint arXiv:2505.24832*.

Moura, Leonardo de, and Sebastian Ullrich. 2021. “The Lean 4 Theorem Prover and Programming Language.” *Automated Deduction – CADE 28*, 625–35.

Nakkiran, Preetum, Gal Kaplun, Yamini Bansal, Tristan Yang, Boaz Barak, and Ilya Sutskever. 2020. “Deep Double Descent: Where Bigger Models and More Data Hurt.” *International Conference on Learning Representations (ICLR)*.

National People’s Congress of the People’s Republic of China. 2021. *Personal Information Protection Law of the People’s Republic of China*.

Nickel, Maximilian, and Douwe Kiela. 2017. “Poincaré Embeddings for Learning Hierarchical Representations.” *Advances in Neural Information Processing Systems (NeurIPS)*.

Noroozizadeh, Shahriar, Vaishnavh Nagarajan, Elan Rosenfeld, and Sanjiv Kumar. 2026. “Deep Sequence Models Tend to Memorize Geometrically; It Is Unclear Why.” *International Conference on Machine Learning (ICML)*.

Oord, Aaron van den, Yazhe Li, and Oriol Vinyals. 2018. “Representation Learning with Contrastive Predictive Coding.” *arXiv Preprint arXiv:1807.03748*.

OpenAI. 2024. “OpenAI O1 System Card.” *arXiv Preprint arXiv:2412.16720*.

Penedo, Guilherme, Hynek Kydliček, Loubna Ben allal, et al. 2024. “The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale.” *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*.

Peters, Ben, Vlad Niculae, and André F. T. Martins. 2019. “Sparse Sequence-to-Sequence Models.” *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 1504–19.

Power, Alethea, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. 2022. “Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets.” *arXiv Preprint arXiv:2201.02177*.

Qwen Team. 2024. “Qwen2.5 Technical Report.” *arXiv Preprint arXiv:2412.15115*.

Radford, Alec, Jeffrey Wu, Rewon Child, oDavid Luan, Dario Amodei, and Ilya Sutskever. 2019. *Language Models Are Unsupervised Multitask Learners*. OpenAI.

Raeesi, Arya, and Hanna Roed. 2026. “Auditing Forgetting in Limited Memory Language Models.” *arXiv Preprint arXiv:2607.00605*.

Ramsauer, Hubert, Bernhard Schöfl, Johannes Lehner, et al. 2021. “Hopfield Networks Is All You Need.” *International Conference on Learning Representations (ICLR)*.

Sakaguchi, Keisuke, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. 2020. “WinoGrande: An Adversarial Winograd Schema Challenge at Scale.” *Proceedings of the AAAI Conference on Artificial Intelligence* 34: 8732–40.

Santos, Saul, Vlad Niculae, Daniel McNamee, and André F. T. Martins. 2024. “Hopfield-Fenchel-Young Networks: A Unified Framework for Associative Memory Retrieval.” *arXiv Preprint arXiv:2411.08590*.

Shazeer, Noam. 2020. “GLU Variants Improve Transformer.” *arXiv Preprint arXiv:2002.05202*.

Shokri, Reza, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. 2017. “Membership Inference Attacks Against Machine Learning Models.” *IEEE Symposium on Security and Privacy (s&p)*, 3–18.

Sondej, Filip, Yushi Yang, and Adam Mahdi. 2026. “RepSelect: Robust LLM Unlearning via Representation Selectivity.” *arXiv Preprint arXiv:2606.17168*.

State of California. 2018. *California Consumer Privacy Act of 2018*. Cal. Civ. Code §§ 1798.100 et seq.

Stojanovski, Zafir, Oliver Stanley, Joe Sharratt, et al. 2025. “Reasoning Gym: Reasoning Environments for Reinforcement Learning with Verifiable Rewards.” *arXiv Preprint arXiv:2505.24760*.

Su, Jianlin, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. 2024. “RoFormer: Enhanced Transformer with Rotary Position Embedding.” *Neurocomputing* 568: 127063.

Talmor, Alon, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. 2019. “CommonsenseQA: A Question Answering Challenge Targeting Commonsense Knowledge.” *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 4149–58.

Team Olmo. 2025. “Olmo 3.” *arXiv Preprint arXiv:2512.13961*.

The mathlib Community. 2020. “The Lean Mathematical Library.” *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP)*, 367–81.

Vaswani, Ashish, Noam Shazeer, Niki Parmar, et al. 2017. “Attention Is All You Need.” *Advances in Neural Information Processing Systems (NeurIPS)* 30.

Wang, Boshi, Xiang Yue, Yu Su, and Huan Sun. 2024. “Grokked Transformers Are Implicit Reasoners: A Mechanistic Journey to the Edge of Generalization.” *Advances in Neural Information Processing Systems (NeurIPS)*.

Wang, Junxiong, Tushaar Gangavarapu, Jing Nathan Yan, and Alexander M. Rush. 2024. “MambaByte: Token-Free Selective State Space Model.” *arXiv Preprint arXiv:2401.13660*.

Wang, Xiaozhi, Tianyu Gao, Zhaocheng Zhu, et al. 2021. “KEPLER: A Unified Model for Knowledge Embedding and Pre-Trained Language Representation.” *Transactions of the Association for Computational Linguistics* 9: 176–94.

Warner, Benjamin, Antoine Chaffin, Benjamin Clavié, et al. 2025. “Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference.” *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*.

Welbl, Johannes, Nelson F. Liu, and Matt Gardner. 2017. “Crowdsourcing Multiple Choice Science Questions.” *Proceedings of the 3rd Workshop on Noisy User-Generated Text (w-NUT)*, 94–106.

Yang, Sohee, Elena Gribovskaya, Nora Kassner, Mor Geva, and Sebastian Riedel. 2024. “Do Large Language Models Latently Perform Multi-Hop Reasoning?” *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.

Ye, Tian, Zicheng Xu, Yuanzhi Li, and Zeyuan Allen-Zhu. 2024. “Physics of Language Models: Part 2.1, Grade-School Math and the Hidden Reasoning Process.” *arXiv Preprint arXiv:2407.20311*.

Zellers, Rowan, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. 2019. “HellaSwag: Can a Machine Really Finish Your Sentence?” *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 4791–800.

Zhang, Biao, and Rico Sennrich. 2019. “Root Mean Square Layer Normalization.” *Advances in Neural Information Processing Systems (NeurIPS)* 32.

Zhang, Ruiqi, Licong Lin, Yu Bai, and Song Mei. 2024. “Negative Preference Optimization: From Catastrophic Collapse to Effective Unlearning.” *arXiv Preprint arXiv:2404.05868*.

Zhao, Linxi, Sofian Zalouk, Christian K. Belardi, et al. 2025. “Pre-Training Limited Memory Language Models with Internal and External Knowledge.” *arXiv Preprint arXiv:2505.15962*.

Zhu, Rui-Jie, Zixuan Wang, Kai Hua, et al. 2025. “Scaling Latent Reasoning via Looped Language Models.” *arXiv Preprint arXiv:2510.25741*.

Formulation Staging

Formulation

Edited:

The repository contains no paper source, so here is the revised LaTeX block:

```
```\latex
\subsection{Facts, storage, and retrieval}
```

A `\emph{fact}` is an atomic (entity, relation, value) triple. In our setting, facts are attributes of individual entities, such as a person's birth city.

We use `\emph{storage}` and `\emph{retrieval}` to describe different abilities. Fact storage means producing a value from the model's weights without any external resource. Fact retrieval means obtaining the value through whatever channel the architecture provides. A dense model can access facts only through its weights. The split model is trained not to store the values directly and instead obtains them from an external store. The architectures therefore differ both in where facts are stored and in how they are accessed.

This distinction creates an attribution problem. A closed-book dense model and a split model with its store attached differ in both storage and retrieval. For example, better performance by the split model could result from reliable access to facts rather than from its parameters learning better reasoning patterns. We therefore include a third condition in which the dense model receives, at evaluation, the same fact values supplied to the split model.

The resulting comparisons answer different questions. Comparing the dense model with and without facts in context estimates the value of making those facts available under the evaluation format. Comparing the split model with the open-book dense model asks whether a difference remains when both models have access to the required facts. This second comparison is the one most relevant to reasoning, although it is not a clean treatment effect unless both models were trained to consume facts in comparable formats. We therefore treat the open-book condition as part of the minimal design rather than as an additional control.

Retrieval by the split model has two stages. `\emph{Addressing}` is the learned stage in which the model emits a query key containing a subject and relation. `\emph{Resolution}` is the store's matching of that key to a row and return of its value. Both stages must succeed for retrieval to succeed.

Each call is allowed to return the value of only one (entity, relation) pair. Otherwise, the store could traverse several relations and perform the composition that we intend to measure in the model. Requiring one fact per call forces the model to construct and issue each successive query itself.

\subsection{Reasoning}

We define reasoning operationally so that it can be measured. Let an \emph{endpoint} be a task used to measure an experimental outcome. Here, a \emph{premise} is a fact or rule supplied in a task item, and the \emph{gold derivation} is the reference sequence of steps needed to reach the answer. An endpoint measures reasoning to the extent that, with fact retrieval and surface form held fixed, it satisfies three conditions:

```
\begin{quote}
\begin{enumerate}
\item \emph{Depth sensitivity}: accuracy declines as the gold derivation
requires more serially dependent steps, or hops.
\item \emph{Distractor invariance}: accuracy changes little when a topically
plausible but logically irrelevant premise is added to the input.
\item \emph{Premise sensitivity}: accuracy falls substantially when a premise
required by the gold derivation is removed.
\end{enumerate}
\end{quote}
```

Each condition addresses a simpler explanation for apparent reasoning. Depth sensitivity tests whether performance depends on the number of required steps rather than only on recalling an answer. Distractor invariance tests whether the model follows the relevant premises instead of reacting to superficial topic overlap. Premise sensitivity tests whether the required premises actually affect the answer. Passing all three provides stronger evidence that the model uses the supplied facts as part of a multi-step derivation rather than merely retrieving them.

We considered two weaker definitions. The first equates reasoning with generalization to held-out items whose answers were absent from training. That criterion also accepts shortcuts. For example, a comparison model might learn from a question template to select the first-mentioned entity without comparing the retrieved values. Such a heuristic could generalize to unseen entities without performing the intended reasoning (<https://arxiv.org/abs/2004.07780>).

The second definition equates reasoning with robustness to perturbations such as added distractors. Robustness is necessary but insufficient. If a model already performs near chance, its accuracy cannot decline enough to make the distractor test informative. This matters here because small language models without post-training, which is outside our scope, commonly perform near chance on multi-step tasks.

For an endpoint to support a claim in either direction, we require four measurement checks:

```
\begin{enumerate}
\item We report chance performance for the answer space of the specific task.
```

For example, chance is  $(50\%)$  for a binary question even if the complete dataset contains many distinct answer strings.

\item We report a \emph{degenerate baseline}: a policy that does not solve the task, such as always predicting the majority class.

\item We report whether the model's output was parsed successfully.

\item We test several plausible phrasings of the same question and report how the score changes.

\end{enumerate}

Without these checks, a null result may reflect a broken or poorly specified measurement rather than the absence of an effect, so we do not draw conclusions from such an endpoint.

When a comparison is conditioned on whether the required facts were available, we determine availability from the corpus and the store rather than from the model's response to a probe. A \emph{probe} is a separate prompt or test used to see whether the model can produce a fact. Selecting items according to whether the model answered such a probe correctly would make the subset depend on the model's own performance and could bias the conditioned comparison.

\subsection{Hypotheses}

Having defined facts and reasoning, we test three main claims in dependency order.

\paragraph{H1: Fact values can be moved out of the weights.}

In the split model, both closed-book recall and the fact content recoverable from its parameters fall to near zero, while its recall with the store attached is at least as good as the dense model's closed-book recall. H1 is a precondition for an interpretable result on H2 and H3.

\paragraph{H2: Moving facts out of the weights does not degrade general language ability.}

Reasoning gains must not come at the expense of the model's other capabilities.

\paragraph{H3: The split model reasons better than its dense twin at matched parameter and token budgets.}

Here, reasoning is measured using the operational criteria defined above.

\paragraph{H0 (the premise behind H3).}

H3 assumes that memorization and reasoning compete for limited capacity, so storing fewer facts frees room to reason. This assumption is central to the memory-split proposal \cite{liemandt} but is not otherwise established. It is observable only if two things hold: the model is near its storage limit (estimated at  $(\approx 2)$  bits per parameter, \citealp{allenzhu2024capacity}), and any drop in stored content is caused by the treatment rather than by how often each fact was seen in training. If the premise fails in the regime we can afford to train in, a null on H3 is evidence about the regime, not the architecture.

\paragraph{H4 (sample efficiency, secondary).}

Freed from memorizing values, the split model reaches a given level of task competence in fewer optimization steps, even if its final accuracy is no higher. This is a claim about learning speed, not final performance.

\paragraph{H5 (behavior on failure, secondary).}

Never trained to produce a value, the split model has no mechanism for inventing one. We therefore expect abstention rather than hallucination on an unanswerable query, and expect the facts available through its store to be directly listable, auditable, and editable without retraining.

...

## Unedited but commented

\subsection{Facts, storage, and retrieval}

A \emph{fact} is an atomic (entity, relation, value) triple. In our setting these are attributes of individual entities, such as a person's birth city.

We separate two properties that are easily conflated. Let \emph{fact storage} be a model's ability to produce a value from the weights alone, with no external resource available and \emph{fact retrieval} the ability to obtain that value by whatever means the architecture provides. For a conventional dense model the two coincide, since the weights are the only channel available. For a model that offloads values to an external store, storage is \emph{theoretically} close to zero by construction while retrieval is supplied from outside. Storage and retrieval are therefore distinct properties, and the two architectures we compare differ on both at once.

[The difference between storage and retrieval can be stated more plainly, like the example I gave later, but other than that, LGTM.]

% Passage is a bit weird.

The distinction limits what a two-condition comparison can show. A closed-book dense model and a memory-enabled split model simultaneously differ in storage \emph{and} retrieval, so a gap between them cannot be strictly attributed to either. [Edit: For example, an increase in reasoning by the split model may be attributed to the benefit of its access to a reliable source of facts rather than to a boost in reasoning due to its parameters encompassing more and better reasoning patterns.] Recovering the two terms requires a third condition, in which a dense model receives at evaluation the same values the store provides to the split model. That gives two contrasts with different meanings[Should this be removed?]. Open-book against closed-book dense measures what having the facts available is worth[This doesn't isolate anything; only the first experiment isolates something.]. Split against open-book dense is the comparison that bears on reasoning, which is why we treat the open-book condition as part of the minimal design rather than as an additional control.

Retrieval in the split model has two stages, which we separate because they generalize differently.

\emph{Addressing} is the model's emission of a query key; \emph{resolution} is the store's lookup of that key. Addressing is the learned stage, but both work in tandem to enable retrieval.

Attributing composition to the model also constrains the store. If a single call can traverse more than one relation, the store performs the composition we would otherwise be crediting to the model, and the comparison measures the retriever. We therefore require that a call return the value for one (entity, relation) pair.

[This makes sense because otherwise the knowledge base would have to encompass the complete answers to the nearly infinite queries a user can ask.]

\subsection{Reasoning}

Treating reasoning as inherent a property of the model invites unresolvable philosophical debate. We instead define it operationally, as a property of measurement.

[Could just delete ``Treating reasoning as an inherent property of the model invites unresolvable philosophical debate" and say, ``We define reasoning as blank to allow for measurement."][disagree]

Let an \textit{endpoint} denote a task used to measure an experimental outcome. An endpoint measures reasoning to the extent that, with fact retrieval and surface form held fixed, it satisfies three conditions:

\begin{quote}

\begin{enumerate}

\item depth sensitivity: accuracy declines as the gold derivation [Edit: What does this mean? Does it mean the specific hops needed?] requires more serially dependent steps;

\item distractor invariance: accuracy is unchanged by adding a premise that is topically plausible but logically irrelevant [Where is this added? What does this mean?];

\item premise sensitivity: accuracy collapses when a premise the derivation depends on is removed. [Same as the point before: what does ``premise" mean in this context?]

\end{enumerate}

\end{quote}

Each condition rules out a specific non-reasoning account of a positive result. Depth sensitivity separates step-by-step derivation from recall of a stored answer; distractor invariance separates reading the premises from matching a surface pattern; and premise sensitivity establishes that the premises are load-bearing rather than decorative. Because none of the three can be produced by fact retrieval alone, the criterion stays meaningful in a study whose only manipulation is how facts are obtained.

[Some of the prose here diminishes reader understanding and could be written more plainly.]

We considered two weaker definitions and rejected both. The first identifies reasoning with generalization to held-out items whose answers do not appear in training; for instance, as estimated through training-set influence. Any task-wide shortcut also satisfies this---a heuristic that happens to generalize is indistinguishable from reasoning under such a definition [Example? What does this mean?] (<https://arxiv.org/abs/2004.07780>)---so it fails to isolate the behavior we care about. The second identifies reasoning with robustness to input perturbation, such as added distractors. Robustness is necessary but not sufficient: a model already near chance has no accuracy left to lose, and small language models without post-training (which we do not perform, and which is outside our scope) typically sit near chance on multi-step tasks, so a distractor test alone cannot certify reasoning. [So the model is already very inaccurate, which is why you can't use robustness, since the accuracy drop-off would not be more telling.]

For an endpoint to support a claim in either direction, we further require that a failure to see an effect be distinguishable from a poorly-defined measurement. Concretely, for each endpoint we state (i) the chance rate for its \emph{specific} answer space, not for the answer pool as a whole [Why do you even need to say ``not for the answer pool as a whole"?]; (ii) the score of a degenerate baseline [What is a degenerate baseline?], such as always predicting the majority class;

(iii) whether the model's output was parsed successfully; and (iv) how the score moves across several plausible phrasings of the same question. An endpoint lacking these cannot separate a genuine null from an instrument failure, and we do not draw conclusions from one. [Good paragraph, but hard to parse.]

A final requirement follows from the distinction between storing a fact and retrieving it. When a comparison is conditioned on whether the required facts were available to a model, that availability must be determined from the corpus and the store, never from either model's behavior on a probe [What does the model's behavior on a probe mean? Does it mean its behavior on a specific prompt or test?]. Conditioning on model output would make the condition depend on the very outcome it is meant to adjust, so the conditioned estimate would no longer be independent of it. [Does this need to be stated?]

\subsection{Hypotheses}

Having defined facts and reasoning, we test three main claims in dependency order.

\paragraph{H1: Fact values can be moved out of the weights.} In the split model, both closed-book recall and the fact content recoverable from its parameters fall to near zero, while its recall with the store attached is at least as good as the dense model's closed-book recall. Note that H1 must be a precondition for an interpretable result on H2 and H3.

\paragraph{H2: Moving facts out of the weights does not degrade general language ability.} Reasoning gains must not come at the expense of the model's other capabilities.

\paragraph{H3: The split model reasons [Maybe add ``according to the definition previously given," or spell it out again.] better than its dense twin at matched parameter and token budgets.}

\paragraph{H0 (the premise behind H3).} H3 assumes that memorization and reasoning compete for limited capacity, so storing fewer facts frees room to reason. This assumption is central to the memory-split proposal \citep{liemandt} but is not otherwise established. It is observable only if two things hold: the model is near its storage limit (estimated at  $\approx 2$  bits per parameter, \citealp{allenzhu2024capacity}), and any drop in stored content is caused by the treatment rather than by how often each fact was seen in training. If the premise fails in the regime we can afford to train in, a null on H3 is evidence about the regime, not the architecture. [Regardless, with the same token count, wouldn't the parameters have room to absorb more information? From the same number of tokens, in one case they may derive more reasoning structure and also an understanding of how to query, while in the other case they have to understand reasoning structure and facts. So the real question is whether it is easier to train a model to query a database to obtain facts and reason with them than to train a model to natively use facts in reasoning.][This is not proven in literature, which some prior editor removed... This presupposes that facts and reasoning compete for space in a meaningful capacity, that reasoning structure isn't more difficult to derive when facts are lacking]

\paragraph{H4 (sample efficiency, secondary).} Freed from memorizing values, the split model reaches a given level of task competence in fewer optimization steps, even where its final accuracy is no higher. This is a claim about the learning curve, not its asymptote [What do you mean by ``its asymptote," and can I just delete ``not its asymptote"?].

\paragraph{H5 (behavior on failure, secondary).} Never trained to produce a value, the split model has no mechanism for inventing one. We therefore expect abstention rather than hallucination on an



unanswerable query, and the knowledge it does use to be enumerable [What does “enumerable” mean in this context?] and editable without retraining.  
...

---

Our main goal is to evaluate the qualitative and quantitative differences between a split-memory LLM and a dense LLM. One of the main claims we want to test is whether a model that does not store facts in its weights will reason better than one that does. Neither half of that sentence has a settled meaning in the literature, so we fix working definitions before describing any experiment. The definitions are narrow on purpose; they are chosen so that each hypothesis below is falsifiable by a measurement we can actually construct—and so that a negative result is informative rather than merely inconclusive.

(what does this mean by neither half?)

→

We seek to evaluate qualitative and quantitative differences between a split-memory LLM and dense LLM across the domains of blank, blank, blank and blank. We define [these terms as blank in order to concentrate and reach insight across these domains fill in]. They are chosen so that each hypothesis below is falsifiable by a measurement we can actually construct

## True Original

```
\section{Formulation}
% \label{sec:definitions}
```

Our overall goal is to evaluate the qualitative and quantitative differences between a split-memory LLM and a dense LLM. One of the main claims we wish to test is whether a model that does not store facts in its weights will reason better than one that does. Neither factual storage nor reasoning has a settled operational meaning in the literature, so in this section we fix working definitions to guide our experiments. These definitions are deliberately narrow: they

are chosen so that each hypothesis can be tested using a measurement we can construct and so that null, inconclusive, and invalid outcomes can be distinguished.

### \subsection{Facts, storage, and retrieval}

A \emph{fact} is an atomic (entity, relation, value) triple. In our setting, facts are attributes of individual entities, such as a person's birth city.

We separate two properties that are easily conflated. Let \emph{fact storage} be the ability to produce a value from the weights alone, with no external resource available and \emph{fact retrieval} the ability to obtain that value by whatever means the architecture provides. For a conventional dense model in the closed-book condition, the two coincide because the weights are the only channel available. For the split model, retrieval is supplied by an external store. Storage and retrieval are therefore distinct properties, and the two systems we compare differ in both.

A closed-book dense model and a store-enabled split model differ in both where facts reside and how they are accessed. If the split model performs better, this may simply be because the store supplies facts that the dense model fails to recall, instead of the idea that the split model reasons better with them. Therefore, we add an open-book dense condition that supplies the same required facts in context. Comparing the two dense conditions shows how performance changes when facts are provided, while comparing split with open-book dense is the more relevant reasoning diagnostic because both have access to those facts. This remains an imperfect comparison because the dense model was not trained to consume facts in the same format.

Retrieval in the split model has two stages. \emph{Addressing} is the model's emission of a query key, and \emph{resolution} is the store's lookup of that key. Addressing is learned by the model, but both stages must succeed for retrieval to succeed.

Attributing \emph{composition}, the combining of two facts in a single step of “reasoning”, to the model also constrains the store. If a single call can traverse more than one relation, the store performs the composition we would otherwise be crediting to the model. We therefore require that each call return the value of only one (entity, relation) pair. This prevents the store from performing multi-hop composition, although it does not by itself establish that the model composes the returned facts.

### \subsection{Reasoning}

Reasoning is difficult to identify from a single accuracy figure. The same number can be produced by a memorized mapping, by a prior over answer formats, or by a scoring harness that fails on one class of outputs. Rather than attempt a general definition of reasoning as an intrinsic property of a model, we define it operationally as a property of measurement. An

endpoint measures reasoning to the extent that, with fact retrieval and surface form held fixed, it satisfies three conditions:

```
\begin{quote}
\begin{enumerate}
\item \emph{Depth sensitivity}: performance is reported across derivation depths and, outside
floor or ceiling regimes, generally declines as the number of serially dependent steps increases.
\item \emph{Distractor invariance}: performance changes little, within some tolerance, when a
(topically plausible but) logically irrelevant premise is added.
\item \emph{Premise sensitivity}: changing a premise required by the derivation changes the
answer, and removing that premise dramatically reduces performance.\end{enumerate}
\end{quote}
```

Each condition eliminates a specific false positive. Depth sensitivity distinguishes derivation from recall of a stored answer, as increased depth indicates a more difficult reasoning task. Distractor invariance tests whether the model uses relevant premises rather than superficial cues. Premise sensitivity establishes that the premises are necessary and sufficient rather than decorative. Combined, these criteria should test for reasoning rather robustly.

We considered a weaker criterion that is insufficient on its own: generalization to held-out items, which does not establish reasoning because shortcuts can also generalize ([\url{https://arxiv.org/abs/2004.07780}](https://arxiv.org/abs/2004.07780)) and reasoning may be (strongly) context-dependent.

Finally, we must distinguish between a reasoning failure due to incorrect fact retrieval from true reasoning failure.

## \subsection{Hypotheses}

Having defined facts and reasoning, we test three main claims in dependency order.

\paragraph{H1: Fact values can be moved out of the weights.}

The split model's closed-book recall and the fact content recoverable from its parameters both fall to near zero, while its recall with the store available is at least comparable to the dense model's closed-book recall. H1 is a precondition rather than a result of interest in itself. If it fails, the treatment did not do what it was designed to do and the remaining hypotheses cannot be evaluated.

\paragraph{H2: Moving facts out of the weights does not degrade general language ability.}

Moving facts to an external store should not reduce performance on tasks that do not depend on those facts. Otherwise, any apparent reasoning benefit may come at the expense of other abilities. We therefore interpret H3 only if the split model preserves its performance on these tasks. An apparent reasoning gain obtained by damaging the model in some other respect is not a gain, so H2 must hold for H3 to be worth interpreting.

\paragraph{H3: The split model reasons better.}

All other factors held constant, the split model outperforms its dense twin on reasoning tasks when both have access to the required facts.

H3 rests on a premise that we state separately because it is taken as fact by Liemandt et al., yet is supported nowhere in the literature [elaboration?]:

\paragraph{H0: competition.}

Memorization and reasoning compete for limited capacity within the model, so decreasing one makes room for the other.

We also record two secondary hypotheses, which are worth testing regardless of H3's results.

\paragraph{H4: Sample efficiency.}

The split model reaches a given level of task competence in fewer optimization steps than the dense model, even when its final accuracy is no higher. This is a claim about the learning curve rather than its asymptote, and it is the form of benefit most relevant to training under a fixed compute budget.

\paragraph{H5: Behavior on failure.}

Because the split model receives no direct supervision to predict stored values, we hypothesize that it fabricates fewer values and produces a failure signal more often when a query cannot be resolved. The external store can be inspected and edited without retraining. Whether the model reliably uses those edits must be tested separately.

# Shared Arch Staging

# Shared Architecture

## Tokenizer and Architecture

Unless otherwise stated, for experimental setup we extended the GPT-2 tokenizer with four control tokens that denote facts (`<|db_start|>`, `<|db_retrieve|>`, `<|db_end|>`, `<|eot|>`). A fact is rendered `<|db_start|>` “name, relation” `<|db_retrieve|>` with loss on followed the value with loss *off*, closed by `<|db_end|>`. Both dense and split models use the same architecture: a Llama-style decoder-only transformer with 12 layers, 12 heads, and d\_model 768 (162 million parameters). They both use pre-norm RMSNorm, RoPE, a SwiGLU MLP, causal attention, and untied embeddings, at a context length of 2048.

## Dense vs Split

We train six models: three dense/split pairs, one pair at each fact load  $N \in \{n50k, n200k, n800k\}$  ( $\approx 0.3\text{M}/1.2\text{M}/4.8\text{M}$  facts). Within each pair, the two models share the same initialization seed, a data-order seed, a step count, and a token budget (3.2B tokens per arm), and they are trained on the same token stream in the same order. In the style of LMLM (Zhao et al. 2025), the sole difference is the treatment of fact values: the **dense** arm sees unannotated inline text and is graded on the value tokens (which it must memorize), while the **split** arm sees the same facts rewritten as a store lookup and has the value tokens masked from the loss (it learns only to issue the query). In this way, we ensure that externalization of facts is the sole manipulated variable. Finally, we compare the two models’ performance on held-out evaluation sets.

## Data Formulation

[Did you guys share the same recipe for synthetic data? Maybe in formulation but Eve, Iris, Stephen all did synthetic data]  
[is this a new part we have to add, i’m not sure where to find this inf]

```
\subsection{Corpus}
\label{sec:corpus}
```

In most of our experiments, our corpus is primarily synthetic; the data are generated in line with existing works [citation]. We do this as we would like to retain precise control over the amount of facts the model is asked to memorize, the predominant independent variable. This also allows us to precisely fix the occurrences of each fact and cleanly separate purely factual and reasoning data.

```
\subsubsection{Entities}
```

We build on top of the robust bioS schema that Allen-Zhu and Li created. To build the corpus, we first start with a list of facts formed by a set of entities. An entity is a synthetic person: a

unique three-word name and a fixed set of attributes drawn uniformly from closed, pairwise-disjoint pools of generated pronounceable words. Closed pools allow for precise measures of information, and although disjointedness is unrealistic, it keeps a correct answer attributable to one attribute.

[might delete] Attribute bits are  $\sum_i \log_2 |P_i|$  per entity; the cost of knowing which  $N$  names exist is a separate term, not folded into the per-entity figure.<sup>footnote{An earlier version of our own accounting folded it in, overstating demand by five orders of magnitude on the experiment's own independent variable.}</sup> Sampling is uniform, which is what makes total demand  $N$  times the per-entity figure: under skew the product no longer holds, and at a fixed exposure budget the tail falls below the floor where capacity is roughly zero, leaving demanded bits unknown.

## \subsection{Rendering and exposure}

Entities render as prose biographies rather than compact records, as the model should still understand natural language. Additionally, [citation] shows that multiple unique exposures of facts disposes them to be learned more robustly by the model. Each rendering draws from 32 templates spanning four length bands, since a single phrasing lets the model store a pattern-to-value association in place of a fact. No attribute appears twice in a document, or effective and nominal exposure counts would differ, and each entity has exactly one surface form, so the model is never trained to recall a canonical name from a variant of it. Each entity appears once per epoch at a position drawn from a per-epoch keyed permutation, which makes exposure counts structurally exact and prevents the model from learning the document sequence rather than the facts.

## \subsubsection{Reasoning slices}

The `\emph{related}` slice asks multi-step questions over the same entities and is where an effect operating through fact access appears first. The `\emph{unrelated}` slice has no dependence on learned facts; degradation there is capacity competition, whereas degradation confined to the related slice is fact access. The unrelated slice must carry no memorizable content of its own or it competes for the capacity being measured, which we enforce by regenerating content per example and refusing a slice whose reuse exceeds a threshold.

[?] The unrelated endpoint is mental modular arithmetic over a flat expression with no chain of thought, chosen for its parameter sensitivity in our range and because having no derivation to parse removes the failure mode that invalidated an earlier arithmetic endpoint of ours. The

related endpoint compares two entities on an ordinal attribute and is answered with the attribute value rather than an entity name, so no copy-from-context policy reaches it.

[delete prob] Floors are measured, not assumed. For each task we search constant answers and copy from every prompt offset over a large sample with a margin for sampling error. Searching constants alone is what costs an endpoint its floor: the comparison task's measured floor fell from 50.2\% to 0.7\% once the answer became the attribute value rather than a name, and excluding zero as a multiplicand moved the arithmetic task's best constant policy from 8.3\% to 6.8\%. An endpoint enters the grid only after sitting away from floor and ceiling in a no-facts control and responding to a sweep of task difficulty. A premise-ablation gate is specified but unbuilt, and fails closed rather than passing by default.

\subsection{Mixture and the demand axis}

Reasoning slices are specified in absolute tokens, never as a fraction; holding the fraction would make reasoning volume move with fact load and confound the result either way. Counts are per slice rather than a shared total, which matters for the controls: a control carries no related slice, so an evenly split total would have given it twice the exposure to the unrelated endpoint as the cells it references, raising its score for that reason alone. Every segment carries a domain token naming its slice, with no option to disable it, because omitting one costs up to twentyfold in capacity at these exposure counts against roughly twofold with one.

Exposures are fixed at 200 in every cell, above the threshold where storage collapsed in our earlier measurements. Fixing them separates fact load from exposure starvation, which an earlier sweep conflated by holding total tokens fixed while raising entity count. Cells are placed by demanded bits per non-embedding parameter, swept geometrically. We also report an oversubscription ratio, but as an interpretive label rather than the placement variable: published capacity measurements bracket our exposure count without covering it, and gated feed-forward layers carry a further penalty, so the assumed capacity is a band and conclusions are checked at both ends. Because the fact slice grows while reasoning slices are fixed, the unrelated slice ranges from about half the smallest cell to a few percent of the largest; its stream is byte-identical across cells, so items truncated at instance boundaries are a uniform tax rather than a per-cell confound.

\subsection{Externalization variants}

The comparison experiments reuse the same entities and rendering, wrapping each fact value in a lookup and excluding those tokens from the loss while grading the query string. The composition variant adds a relation between entities and renders two-hop items with an explicit derivation, so the second query must carry the bridge name returned by the first. Entities split into a population



appearing in composed items and one appearing only in single-hop biographies, giving a held-out test of the composition procedure over facts the model stores but has never composed. The crowding grid carries no masked arm, since it asks whether storing facts costs reasoning in a single dense model and a second arm would confound that with the intervention; the data path reserves an optional per-instance label mask so the arm can be added later.

# Standard Regime Staging (Exp 1)

# Facts and Reasoning under a Standard Training Regime

## Research Question and Hypothesis

Does MemorySplit improve reasoning under a standard training regime? MemorySplit relies on three premises:

1. arbitrary facts occupy parameters in a small model
2. masking those facts frees those parameters
3. the freed parameters get taken up by reasoning

If all three hold, the split arm's advantage over its dense twin on held-out reasoning should widen as fact load rises, since the dense arm is most constrained at high load. We test this as a difference-in-differences between the lightest and heaviest loads, against the null that the advantage does not vary with load.

## Experimental Setup and Dataset

We train two models that are identical in initialization, data order, schedule, hyperparameters, and token budget. They differ in whether the loss rewards predicting a fact value. Values remain visible in the context and in an external store throughout, so the split arm can read any fact without being asked to learn it. The masking costs no additional training FLOPs.

We generate the corpus ourselves, so the fact load is a controlled dose with a known information content, and the reasoning tasks are knowledge free by construction. The corpus is 54 percent FineWeb-Edu bed, 23 percent biographies, 12 percent iGSM, 8 percent deduction, and 3 percent fact-use QA. The biographies follow the bioS protocol (Allen-Zhu and Li, 2023) and supply the fact values that are masked from the split arm's loss and served from the external store.

Reasoning tasks come from two generators, a RuleTaker-style deduction generator at depth one to two (Clark et al., 2020) and a reduced iGSM generator, and every task is checked against an oracle that re-derives its answer.

We ran our setup at two scales. The 160M sweep crosses three fact loads with two arms at 3.2B tokens per run, which is roughly 20 tokens per parameter and therefore compute optimal.

Biographies are about 23 percent of a 3.2B-token mixture, so the biography budget is fixed at roughly 732M tokens. The 1B pair on AWS serves as a scaling confirmation, as we have one seed pair there due to compute constraints.

## Validation

The split model is able to learn to offload facts and perform queries. At 50K entities, 94.9 percent of its queries resolve to a fact present in the store, and disabling the store drives closed-book recall to zero. It generalizes to entities it never read, provided the query takes a form it has seen; in a 0.8B-token pilot at the 200,000-entity load it scores 0.996 on fresh-entity lookup with the store attached and 0.000 with it detached, against 0.002 to 0.008 for the dense arms across the three loads ( $n = 500$  per cell). The corresponding knowledge in the dense arm is spatially localized: ablating its 64 most fact-selective units cuts closed-book recall from 0.66 to 0.35, a 47 percent reduction, against 3 percent for an equal number of random units, and at the same indices in the split arm selectivity is three to six times lower,  $|d|$  of 1.5 to 3 against 7 to 11..

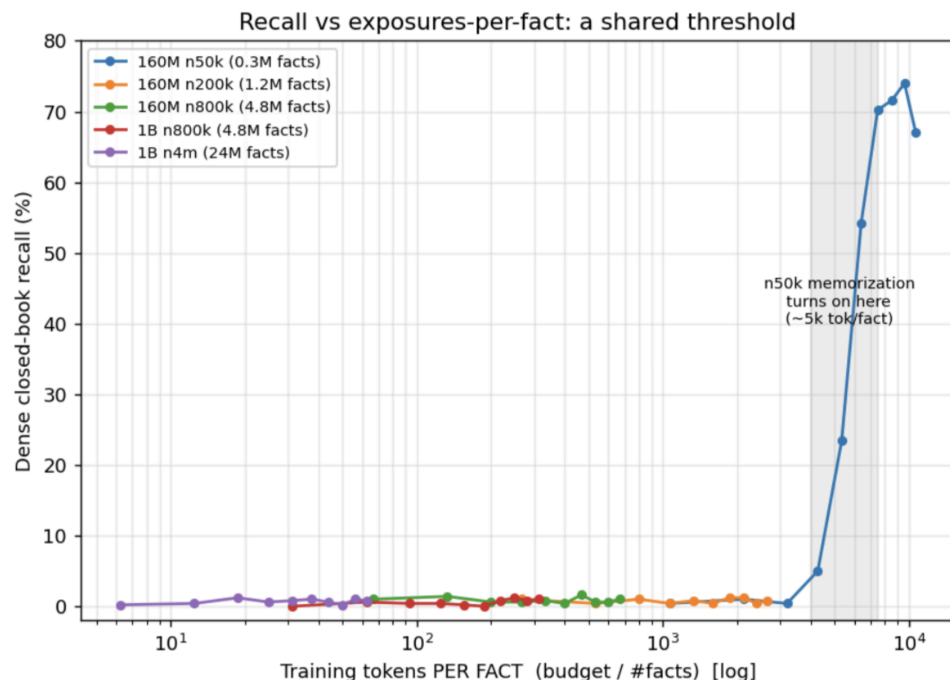
## Results

Under a standard training budget, split trivially does not improve reasoning, because the dense model seems to never overcrowd.

On held out deduction, the split arm scores 2.0 points below its dense twin at 50k entities, 3.1 above at 200k and 5.5 below at 800k. There is no consistent advantage, and the sign reverses across the three loads. The hypothesis predicts that the advantage should grow as fact load grows, since that is when the dense twin should be most burdened, and instead the difference-in-differences comes out at negative 3.5 points.

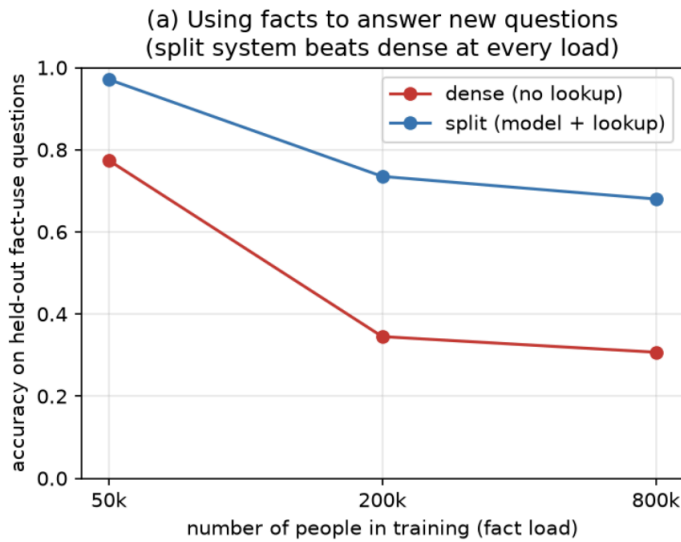
Further investigation shows that the dense model's reasoning does not go down as it is shown more facts. Going from 50k to 800k entities multiplies the fact load by sixteen, and dense reasoning accuracy moves from 65.0 to 66.7 to 68.2, so it does not fall. The crowding account predicts degradation, but reasoning is flat at worst, discrediting the crowding account. We assumed a model shown too many facts would compress them until something gave way, whereas what it actually seems to do is decline to learn them, and that costs it nothing.

The storage measurements explain why we see these results. At 50k entities, the dense model holds around 33 of the roughly 53 bits per entity that we showed it, but at 200k and 800k it holds about 0.2 bits per entity, which is close to nothing. This is because, given the fixed token budget of 732M tokens, the number of exposures falls from about 196 exposures to 49 and then 12 as the number of facts rises. As seen in Figure [#], the dense model only recalls facts when there are enough training tokens per fact.



Nothing in our sweep came to the capacity that crowding would require. Allen-Zhu and Li (2024) measure achievable storage at two bits per parameter, giving a ceiling of 324M bits for our 162,220,800-parameter model, and each entity in our corpus carries roughly 53 bits by construction. At 800,000 entities the demanded load is 42M bits, or 13 percent of that ceiling. Their two-bit figure is measured at 1,000 exposures per fact, however, and our budget allows only 12 to 196. At 100 exposures the same work reports one bit per parameter, which halves the applicable ceiling and places even our heaviest load at 26 percent of capacity.

## Other results



---

Ask a dense model about a person not in training data and it hallucinates a wrong answer about 99% of the time. Conversely, the split model does that essentially never (0% at n50k/n200k, 0.08% at n800k). Across 5,000 tries it also never once returned a real fact about the wrong person. Instead of guessing it **re-runs the lookup or trails off without naming a value.**

# Reasoning Staging (exp 2)



# Two-Hop Fact-Based Reasoning (done in main doc)

## Math Fact-Based Reasoning

### Research Question and Hypothesis

While two-hop reasoning is fact-based, our implementation remains reducible to a fact-agnostic solving recipe, as it is the deterministic composition of two sequential queries. As such, we intend to evaluate a stronger version of the same question regarding whether a split model can perform fact-based reasoning: by being responsible for mathematical reasoning from a given block of premises. Unlike two-hop (which has a fact-invariant general approach), we theorize that the path taken to do a mathematical proof notably changes on the basis of the axiomatic “facts” at hand. By making the problem space contingent on facts, the model will be more directly tested on fact-based reasoning in the stricter sense, rather than as more generally understood. Furthermore, unlike previous experiments which attempt to implement a retriever system to aid the model, we instead assume a “perfect” retriever which, unlike LMLM or Co-LMLM, doesn’t even need to be queried to know exactly what facts are required to solve the problem. By doing so, the problem of retriever optimality can be abstracted away and the pure viability of offloading facts to an entirely separate system from the LLM can be evaluated (Zhao et al., 2025, Feldman et al., 2026).

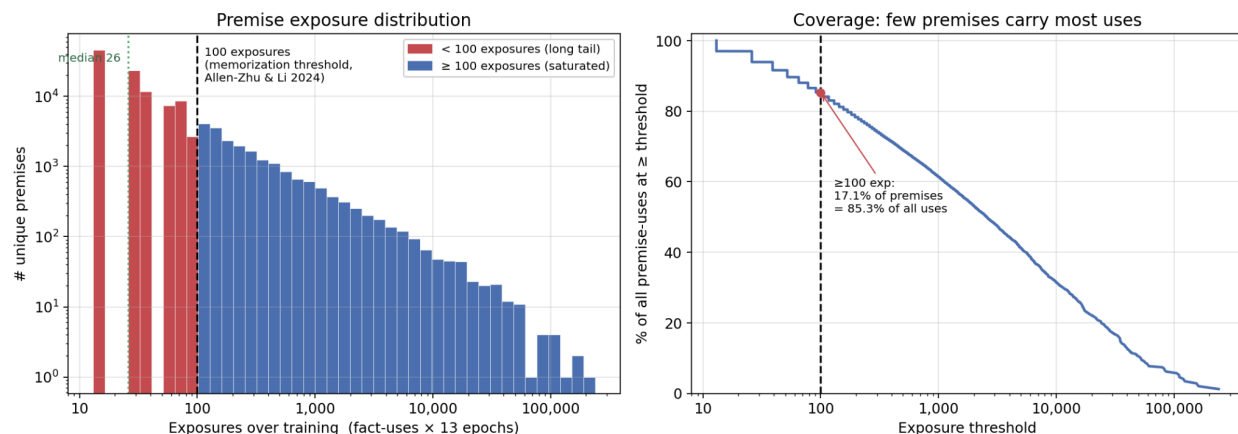
### Experimental Setup and Dataset

To successfully maintain a discriminant between the split and dense model, instead of pre-training a model from scratch and risking both the split and dense model achieving 0% on the mathematical problems from poor initial generalization, we instead take the final trained checkpoint of Qwen2.5-0.5B base, which successfully generated ~50% (NOTE: ask Tejas for the exact stat) of next tokens and 0% of entire valid proofs under teacher forcing and then do supervised fine-tuning of both the split and dense models. Qwen 2.5-0.5B is already trained on some mathematical reasoning, which means that both the split and dense model technically have some mathematical facts already, but given the largely synthetic naming scheme of proofs and relatively low preexisting success rate this was an accepted point of potential further clarification.

The dataset was tokenized using the Qwen2.5 tokenizer for maximum compatibility. Mathematically formal proofs were compiled from Metamath’s set.mm, iset.mm, and nf.mm, Mizar, Magnushammer (a solved set from Isabelle), thproofs, ATP/E-prover prf2, and distinct Mizar alternate runs from ENIGMA mzzr 01/02/03/08 for a total of 467.2 million tokens, repeated

for 13 epochs in order to achieve a total of 6.07 billion tokens seen by the model. 17.08% of unique premises receive 100 exposures, which is by Knowledge Parameter Scaling Laws sufficient for somewhat memorizing the premises, for a total of 85.28% of premise uses in the data (Allen-Zhu and Li, 2024). Our math proofs are organized as premise blocks starting with “I know these mathematical statements:” followed by the list of factual premises (as well as a list of local assumptions), and then a “----” to indicate the start of the actual proof, which has starts with a supervised “GOAL” line that the split and dense models target.

P3 formal-proof corpus (v3): 120,377 unique premises, 1,517,520 premise-uses, 13 epochs



Conceptually, the setup during training is similar to Co-LMLM by wrapping a fact by a special query sequence, but unlike Co-LMLM the split model the wrapper around the factual premise block is also masked out in Co-LMLM style, as it is assumed that the split model doesn’t even need to know what or when to query for the perfect retriever to gather and return all necessary info to solve the problem (Feldman et al., 2026). Dense, by contrast, also receives the trace with facts provided but does not mask them out, instead including the facts in the loss function and encouraging memorization. During gradient backpropagation, rather than dividing the total gradient by the number of supervised tokens (which might fairly be interpreted as equivalent to increasing learning rate for the reasoning section), we instead divide by a fixed 262, 144 constant to represent the global batch size (which is the number of supervised tokens per batch for dense and number of total tokens per batch for split). Through this, essentially only the reasoning section of the gradient is preserved at similar ratios to the reasoning section of the dense, rather than being multiplied. While multiplying would likely benefit split, the goal was to isolate the “noise,” so to speak, created by tuning for data.

## Evaluations

During training, cross-entropy loss and supervised token fraction were monitored. The split model has loss on fewer tokens than the dense model which is also trained to memorize the fact block, so loss was expected to be lower for the split model without any particular meaning.

During the evaluation runs, a held out set of proofs, `facts_present`, from each family were used with facts in the premise block that the models have never seen before anywhere in a mathematical trace either as premises, intermediates, or the goal. Both models are run over the entire set of four thousand held out traces with the premise block given. Additionally, for a deterministic 10% of reasoning traces (`facts_absent`) the two models must also try to solve it without being given a premise block to verify the split does not learn any facts and the dense model learns facts, and for 10% (`facts_corruption`) statements of premises are changed to measure split and dense generalization. For all three categories `facts_present`, `facts_absent`, and `facts_corruption`, cross-entropy loss and token prediction accuracy is measured using teacher forcing. The Metamath proofs for `facts_present` are also all deterministically verified using the Metamath tri-state solver, which categorizes them into valid, invalid, or unknown. For the run with the premise block, teacher forcing is used to give a percent of tokens generated correctly, and for the metamath sub-section of the eval the proof generated by the split and dense models is also deterministically verified. There are 4191 eval rows used for a total of 7.041 million tokens (8.46 million including the 10% diagnostics) and Qwen2.5-0.5B base is run on the same eval set as a base benchmark.

## Results

When the correct premise block is given to both models, the split model performs better than dense on every end-to-end Metamath endpoint, while Qwen2.50.5B base does not generate a successful proof under any of them. Furthermore, the confidence intervals for each splitdense difference remain entirely above zero, suggesting that the advantage is not dependent on a small number of unusually easy proofs for split. While the absolute amount of wholeproof success remains relatively low, the agreement between exact match and deterministic validity means the difference cannot be explained only by the split model producing a different but still invalid proof form. As such, when access to factual premises is held equal, removing the responsibility of memorizing those premises does not harm mathematical reasoning and instead appears to somewhat improve it

Endpoint	base	dense	split	$\Delta$ split–dense [95% CI]
Exact match (evaluated)	0.00%	7.89%	10.93%	+3.04% [+1.21, +5.06]
Exact match (budget-eligible)	0.00%	8.44%	11.69%	+3.25% [+1.30, +5.41]
Metamath validity (decided)	0.00%	9.11%	12.35%	+3.24% [+1.21, +5.26]

`facts_absent` (header only) — split – dense:

Family	n	Base NLL (acc)	Dense NLL (acc)	Split NLL (acc)	$\Delta$ NLL split–dense [95% CI]	$\Delta$ Acc split–dense [95% CI]	Gold paste
enigma	27	0.3080 (91.29%)	0.0780 (97.84%)	0.0835 (97.88%)	+0.0055 [+0.0036, +0.0077]	+0.04% [−0.03, +0.10] (CI incl. 0)	76.2%
isabelle	59	0.7679 (83.78%)	0.4565 (88.92%)	0.4462 (89.39%)	−0.0103 [−0.0206, −0.0002]	+0.46% [+0.20, +0.80]	72.4%
metamath	50	0.5020 (88.87%)	0.2910 (93.22%)	0.3720 (92.38%)	+0.0811 [+0.0676, +0.0987]	−0.83% [−1.03, −0.69]	78.6%
mizar	249	1.0685 (73.57%)	0.7197 (80.06%)	0.7527 (79.86%)	+0.0330 [+0.0256, +0.0421]	−0.20% [−0.33, −0.07]	29.3%
prf2	32	0.3312 (90.59%)	0.0883 (97.60%)	0.0972 (97.56%)	+0.0089 [+0.0064, +0.0118]	−0.04% [−0.10, +0.01] (CI incl. 0)	77.8%
thproofs	5	0.9615 (78.39%)	0.6666 (83.80%)	0.6850 (84.20%)	+0.0184 [+0.0061, +0.1249]	+0.40% [−0.27, +0.68] (CI incl. 0)	21.7%

When the premise block is removed, the direction of the result generally reverses, with split receiving a worse likelihood than dense for nearly every family and the strongest loss in next token accuracy occurring on Metamath. This is the behavior that should occur if dense has stored part of the premise content in its weights while split has learned to depend more heavily on the externally supplied facts. The Isabelle exception, as well as accuracy differences that are close to zero for several families, means this should not be interpreted to show that split learns no mathematical facts at all. Instead, when considered alongside the `facts_present` condition, the result demonstrates that a greater portion of split’s performance is contingent on access to the premise block, and the base Qwen model likely has some pre-existing knowledge of the facts in the dataset.

facts\_corrupted (names kept, statements swapped) — split – dense:

Family	n	Base NLL (acc)	Dense NLL (acc)	Split NLL (acc)	$\Delta$ NLL split–dense [95% CI]	$\Delta$ Acc split–dense [95% CI]
enigma	27	0.3446 (90.29%)	0.0793 (97.69%)	0.0883 (97.71%)	+0.0090 [+0.0062, +0.0122]	+0.02% [−0.06, +0.09] (CI incl. 0)
isabelle	59	1.0054 (79.37%)	0.6061 (85.75%)	0.6105 (85.69%)	+0.0044 [−0.0061, +0.0189] (CI incl. 0)	−0.05% [−0.28, +0.18] (CI incl. 0)
metamath	50	0.3741 (90.43%)	0.2533 (93.63%)	0.3074 (92.88%)	+0.0541 [+0.0459, +0.0648]	−0.76% [−0.93, −0.62]
mizar	249	1.0065 (75.12%)	0.6111 (82.66%)	0.6146 (82.80%)	+0.0035 [+0.0007, +0.0066]	+0.14% [+0.05, +0.24]
prf2	32	0.3186 (90.69%)	0.0795 (97.62%)	0.0851 (97.70%)	+0.0057 [+0.0035, +0.0082]	+0.08% [+0.03, +0.12]
thproofs	5	1.3109 (69.77%)	0.7563 (80.18%)	0.7896 (80.23%)	+0.0332 [+0.0102, +0.0686]	+0.05% [−0.59, +0.65] (CI incl. 0)

# Info Saturation Staging (exp 3)

# Reasoning under Information Saturation

## Research Question and Hypothesis

Text

## Experimental Setup and Dataset

Text

## Evaluations

Text

## Results

Text

Model	N at $\rho=1$	$\rho=0.25$	$\rho=0.5$	$\rho=1$	$\rho=2$	$\rho=4$
15M	378k	0.3h	0.5h	1.1h	2.1h	4.2h
30M	756k	1.1h	2.1h	4.2h	8.4h	16.8h
60M	1.51M	4.2h	8.4h	16.8h	33.6h	67.2h
120M	3.03M	—	—	67.2h	134.5h	—

Component	B200-h, 1 seed	B200-h, 3 seeds
15M/30M/60M grid, 15 cells	171	513
120M at $\rho=1$ and $\rho=2$	202	605
Checkpoint axis, 70M $\times$ 3	69	206
Zipf cell at $\rho=2$	8	25
Reasoning-only controls	3	9
<b>Total</b>	<b>~453</b>	<b>~1,360</b>

Demanded b/p	Occupancy	Predicted R(F)	Predicted recall
0.5	25%	0.50	~100%
1.0	50%	1.00	~100%
1.8	90%	1.80	~100% (hinge)
2.5	125%	2.00	~80%
3.0	150%	2.00	~67%
4.0	200%	2.00	~50%

But the hinge recall should fall as capacity/demand. If we instead see a smooth R



# Info Saturation Co-LMLM Based

### Hypothesis and Setup:

Our primary focus is to answer the hypothesis that preventing the memorization of facts in a language model’s weights would allow it to reallocate those parameters for better reasoning. This experiment takes a similar approach to Co-LMLM with regards to preventing model memorization of facts.

### Data:

We used the Co-LMLM’s data annotation method. We first used a frontier model to identify fact spans in 45k FineWeb-Edu documents using the annotation prompt from Co-LMLM and then finetuned our own modernBERT decoder model to make the same fact span identifications. We then ran the modernBERT annotator model over 750M FineWeb-Edu tokens to create our corpus. The fact spans make up ~7% of the total tokens in the corpus. As defined in Co-LMLM, “a fact span is a sub-string of a pre-training document  $x$  that encodes world knowledge we want our model to externalize”. These fact spans are usually short, self-contained bits of exact information like “Lycopene is a member of the <Fact>carotenoid family</Fact>”.

### Experiment:

We ran two models, a base model and a split model at 135M parameters with the SmolLM2 architecture. The base model was trained normally over the corpus. The split model was trained where the identified fact spans had their loss masked out so that the model would not be trained to predict them and thus not memorize them in its weights.

It is important to note that although we use several ideas about masking facts from Co-LMLM we do not include their retrieval system. To isolate any reasoning change from fact masking, we do not include the extra burden of learning to generate queries to access a knowledge base like in Co-LMLM.

We then train each model for 20B tokens over the corpus (~26.7 epochs) and we monitor the model’s performance on evaluations to watch for overfitting. We do this overtraining so that we can get higher fact exposures which encourages the base model to memorize more facts.

### Evals:

For the evaluations, we run CSQA, HellaSwag, PIQA, SIQA, ARC-Easy for reasoning evaluations and TriviaQA and PopQA for factual knowledge evaluations.

### Results:

Plateau Average of last 10 checkpoints

Model	CSQA	HellaSwag	PIQA	SIQA	ARC-e	Avg
Base	26.54 $\pm$ 0.97	31.24 $\pm$ 0.12	63.66 $\pm$ 0.44	40.81 $\pm$ 0.25	53.26 $\pm$ 0.55	43.10 $\pm$ 0.26

Split	27.27 $\pm$ 0.87	31.33 $\pm$ 0.16	63.08 $\pm$ 0.39	40.70 $\pm$ 0.27	52.23 $\pm$ 0.65	42.92 $\pm$ 0.20
-------	------------------	------------------	------------------	------------------	------------------	------------------

Model	TriviaQA	PopQA
Base	7.13 $\pm$ 0.26	7.71 $\pm$ 0.47
Split	6.06 $\pm$ 0.20	6.92 $\pm$ 0.42

### Conclusions:

The base model was consistently higher than the split model on fact recall while both models performed the same on reasoning. The lowered fact recall of the split demonstrates the lowered memorization of facts in the split model due to fact masking during training. Since the reasoning ability of the models stayed the same, the split model did not gain an advantage over the base model from less stored factual information. This is likely due to the fact spans usually being small self contained pieces of knowledge like names or dates instead of the dense relations of ideas and reasoning that are far more difficult to isolate and mask. For example, with our annotation method, we may identify and mask out the names “Adam Smith” and “Wealth of Nations” but not their ideas of economics and free markets. Therefore, the split model still is trained to memorize the economic ideas and reasoning of Adam Smith and Wealth of Nations but not have to memorize their names. Since the split model knows less but the reasoning ability remains the same, it indicates that these pieces of information likely only make up a small component of the total information a model memorizes and thus masking them out does not free up a significant portion of model weights to be used elsewhere.

### Citations:

<https://arxiv.org/abs/2607.07707>

## Conclusion Staging (Leave)

In this paper we have run a battery of experiments to evaluate the potential advantages and disadvantages of a split-memory system compared to a dense language model. Blah blah.

Overall, our results seem to indicate that split models tend not to degrade significantly in performance (natural language and reasoning) relative to dense models, but they do not perform better, either. Split models may also converge on less compute, as they do not need to spend as much time or as many tokens memorizing facts. Similarly, since the memory store is independently updatable and replaceable, it may proffer advantages for quickly modifying a LM's knowledge without expensive retraining. Conversely, the method and content of the store may require specialized upkeep and independent verification, which could negate the above advantages.

We have also seen that the actual design of the memory store has an outsized impact on the model's performance; the memory store cannot simply be abstracted away and must be engineered in consideration with the rest of the model design. (for example modern hopfield, cosine similarity, exact key-value lookup, fuzzy key-value, continuous lookup, etc).

smth smth comment on generalization to OOD or not, fact recall, hallucination/forgetting.

Future work? post-training + RL (likely improves stuff a lot a lot). blah blah

---

## Implications

\subsection{Implications for the Field}

Our results show that fact values can be removed from model weights without measurably harming the tested language and reasoning abilities. The split model performs better on fact-based tasks because it retains almost no fact values and retrieves them from the store which provides a more reliable access to the required facts. Factual access therefore does not require factual memorization.

Moreover, our dense models never approached their storage capacity: when facts received too few training exposures, the models simply did not memorize them. A stronger test must expose many more facts often enough to create genuine competition for model capacity.

This result complements Engram \citep{cheng2026engram}. Engram offloads frequently used linguistic patterns and reports reasoning gains, whereas our architecture offloads explicit entity-level facts. Together, these results suggest that external memory is most useful for information the model would otherwise learn and process repeatedly. Rarely seen facts may offer less benefit because the model often does not memorize them.

Externalization still offers benefits. Facts can be inspected, replaced, or deleted in the store without retraining the model, and the learned query interface may reduce the training needed for

new fact-based tasks. However, these benefits depend on reliable addressing and store maintenance. Behavioral improvements after edits, unlearning, and reduced hallucination remain to be tested directly.

## Limitations and Future Experiments

Most of our data was collected with one-seed runs, which means that it points towards directional indicators for improvement, rather than decisively clarifying the exact impact of our targeted changes. In the future, carrying out more seed runs with stronger controls could demonstrate the more exact impact of the memory split model.

The majority of our work consists of a single seed run

- Some legacy reasoning endpoints performed near or below non-reasoning baselines.
- Weight probes can show no detectable factual information, but cannot prove that none exists.

The evidence is stronger for factual externalization than for improved reasoning.

- The experiments did not create enough factual occupancy to test whether facts crowd reasoning out of model weights.

Entity count and exposures per fact were confounded in the 160M sweep.

Seed limited

- Three seeds were insufficient to rule out reasoning effects of several percentage points.
- Several mechanistic and two-hop results relied on only one seed.

## Future Work

Run a capacity-occupancy sweep while holding exposures per fact fixed.

Evaluate more model sizes and seeds to distinguish capacity effects from optimization effects.

Train Dense and Split to consume the same factual interface.

Evaluate stronger reasoning tasks with novel facts and controlled counterfactuals.

Measure end-to-end compute, latency, storage, and maintenance costs.